MSc in Mathematics Dissertation: Hyperbolic Geometry and Sandpiles

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Introduction

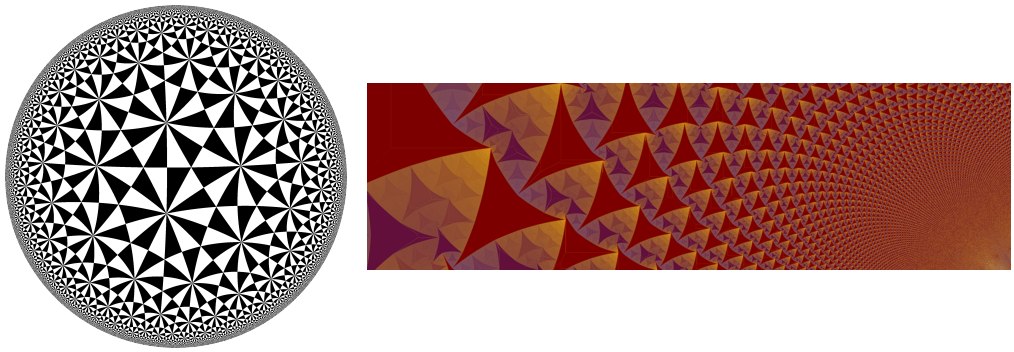


Figure 1: *Left*: A tiling of the Poincaré disc, one of the models of hyperbolic space. *Taken from Wikipedia. Right*: A close-up of a sandpile used in an article which interestingly compares the sudden coordinated motion of sandpiles to political phase transitions such as the Arab Spring or the collapse of the Eastern Bloc [1].

From the start of this project a goal was to see how Abelian sandpiles might react to being imposed upon graphs in and tilings of hyperbolic space. In layman's terms we wanted to mash together the two pictures in Figure 1. This was not entirely achieved in general terms but it was in one specific scenario. Rather than diving head-first into the deep end there is a good deal of work needed to be done in order to reach this point safely.

The first chapter introduces the basics of hyperbolic geometry with help from [2] and [3]. This starts by defining the upper half-plane model describing notions of length and distance and how they differ from the more familiar Euclidean geometry. As the chapter progresses the shortest paths between two points, commonly known as geodesics, are considered in the both the

upper half-plane model and another, the Poincaré disc model.

Introducing these ideas of a new geometry requires a lot of formal definitions, propositions, and theorems. Hopefully the inclusion of diagrams, examples, and asides makes the reading of these a less arduous task and provides a more holistic view of the approach taken in writing this paper.

The second chapter begins by presenting the idea of chip-firing on graphs with vertices and edges as described in [4] and builds up to trying to describe the structure of a special system which uses these graphs in relation to some tilings of hyperbolic space.

In chip-firing each vertex of a graph is assigned an integer value (of chips) and when a vertex is selected to be fired it sends one chip along each outgoing edge to all adjacent vertices. This means that its value will decrease by its degree and all of the neighbouring vertices will increase in value by one.

One place where chip-firing plays an essential role is in that of Abelian sandpile models and, unsurprisingly, this is the model we will be interested in. Here the values assigned are non-negative and there is one globally accessible vertex designated as a sink where all chips vanish. All of the other vertices are fired until the amount of chips - sometimes thought of as grains of sand, hence the name - they have lies between zero and the degree of the vertex. When placing large numbers of chips on grid graphs and allowing them to stabilise we often discover that interesting fractal patterns are the result because of a property called self-organised criticality.

The simplest cases of hyperbolic tilings of the Poincaré disc to work with in this sense are the ideal tilings where all vertices of shapes lie on the boundary because the dual graphs of these are regular infinite trees. Being able to quickly - i.e. without having to perform an eye-watering number of firing moves or any at all for that matter - determine the stable state of the regular infinite tree where all vertices have a degree of four after dropping any number of chips we like on the vertex declared to be at the centre was achieved but this is as far as things got.

Another of the ambitious and ultimately a little out of reach goals set at the beginning of this project was to create some sort of immersive webpage where a user can interact with an infinite hyperbolic tiling - something such as being able to navigate around in the tiling and inspect how things look close to the boundary of the Poincaré disc in order to see some of the (potentially

fractal) patterns. So as I was writing this paper I was simultaneously writing code.

It soon became clear that the best language to use to be able to make an interactive webpage such as the one I had in mind was JavaScript and so the first step I had to take was to actually learn some JavaScript. After taking a pair of basic online courses offered by Codecademy ([5] and [6]) I felt ready to go.

The first idea I had was to construct something where a user could input points in hyperbolic space and have the geodesics between them drawn in return. Then a page where the user can click on and drag points around the half-plane and see how the geodesic changes and a page where two points can be transformed by any of a special group of modifications known as Möbius transformations. The last page made was one where a user can choose to place up to 159 chips on the central vertex of the four-valent infinite tree and receive the relevant coloured tiling of the Poincaré disc in return.

In the appendix at the end of this paper there is more detailed discussion of how I managed to do this along with snippets of the code itself and some screenshots of examples displaying what is produced by the different pages of the website.

Chapter 1

Hyperbolic geometry

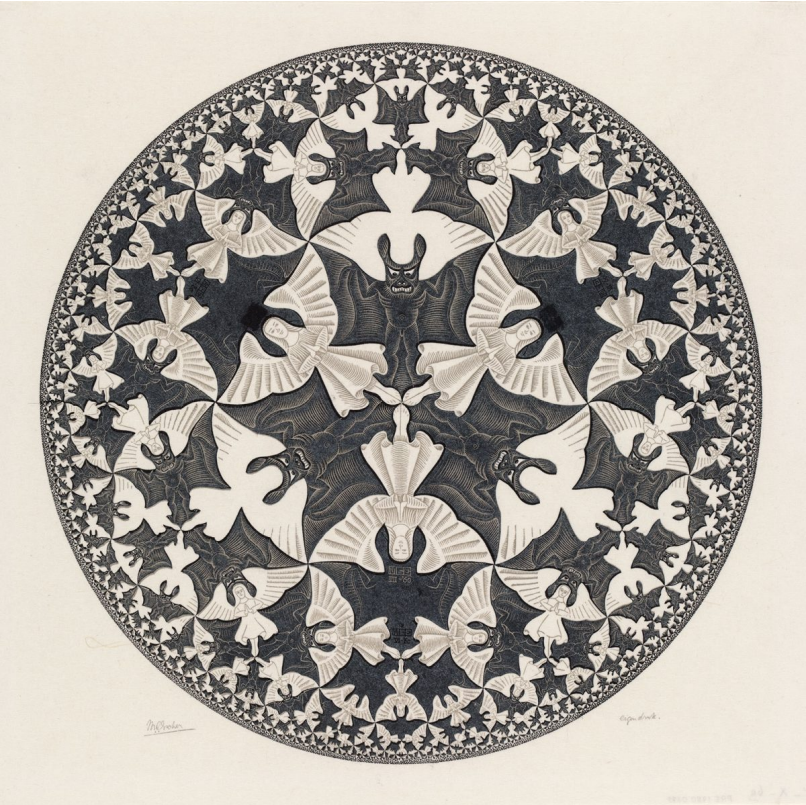


Figure 1.1: Circle Limit IV (Heaven and Hell), 1960 by M. C. Escher.

The five postulates defining the basic rules of plane geometry given near the beginning of the first book of Euclid's *Elements* are as follows:

1. A straight line segment may be drawn from any given point to any other.
2. A straight line may be extended to any finite length.
3. A circle may be described with any given point as its centre and any distance as its radius.
4. All right angles are equal to one another.
5. If a straight line intersects two other straight lines, and so makes the two interior angles on one side of it together less than two right angles, then the other straight lines will meet at a point if extended far enough on the side on which the angles are less than two right angles.

Along with the five axioms and twenty-three definitions they are presented beside these form the basis for the extensive proofs in the text.

Hyperbolic geometry is a non-Euclidean geometry that rejects the fifth postulate, often referred to as the parallel postulate. Euclid's other four postulates are admitted by the rules of hyperbolic geometry and so many of the theorems are identical to those in Euclidean geometry. Some do differ though, giving us something to talk about.

The first published works on hyperbolic geometry [7] were by the Russian mathematician and geometer Nikolai Lobachevsky (1792 - 1856). Owing to the revolutionary character of his work Lobachevsky has been referred to by some as the Copernicus of Geometry [8] and in some texts hyperbolic geometry is actually called Lobachevskian geometry.

1.1 The upper half-plane model

We will begin our exploration by first defining one of the most used models of two-dimensional hyperbolic geometry, the upper half-plane.

Definition 1.1.1. The *upper half-plane* $\mathbb{H}$ is the set of complex numbers z with positive imaginary part: $\mathbb{H} = \{z \in \mathbb{C} : \mathrm{Im}(z) > 0\}$.

Definition 1.1.2. The *circle at infinity* or *boundary* of $\mathbb{H}$ is defined to be the set $\partial\mathbb{H} = \{z \in \mathbb{C} : \mathrm{Im}(z) = 0\} \cup \{\infty\}$. That is, $\partial\mathbb{H}$ is the real axis together with the point ∞.

Equipping this with the metric

$$ds^2 = \frac{dx^2 + dy^2}{y^2}$$

gives us a model of the hyperbolic plane.

1.1.1 Length and distance

Now that we have a model to work within we can introduce the relevant notions of length and distance.

A path σ in the complex plane $\mathbb{C}$ is the image of a continuous function $\sigma(\cdot) : [a, b] \to \mathbb{C}$ where $[a, b] \subset \mathbb{R}$ is an interval. We will assume that σ is differentiable and that the derivative σ' is continuous.

Definition 1.1.3. Let $\sigma : [a, b] \to \mathbb{H}$ be a path in the upper half-plane $\mathbb{H}$. Then the *hyperbolic length of* σ is obtained by integrating the function $f(z) = \frac{1}{\mathrm{Im}(z)}$ along σ, i.e.,

$$\mathrm{length}_{\mathbb{H}}(\sigma) = \int_\sigma \frac{1}{\mathrm{Im}(z)} dz = \int_a^b \frac{|\sigma'(t)|}{\mathrm{Im}(\sigma(t))} dt.$$

Now we know how to find the total length of any given path between two points in the upper half-plane which satisfies our conditions pertaining to differentiability and continuity. The next thing we might be interested in knowing - which is often more desirable - is how to find the length of the shortest path between any two points.

Definition 1.1.4. Let $z, z' \in \mathbb{H}$. We define the *hyperbolic distance* $d_{\mathbb{H}}(z, z')$ between z and z' to be

$$d_{\mathbb{H}}(z, z') = \inf\{\mathrm{length}_{\mathbb{H}}(\sigma) : \sigma \text{ is a piecewise continuously differentiable path with end-points } z \text{ and } z'\}.$$

Thus we consider all piecewise continuously differentiable paths between z and z', calculate the hyperbolic length of each such path, and then take

the shortest. This infimum is achieved by a path known as a geodesic. In both Euclidean and hyperbolic geometry geodesics are unique but in general they are not unique elsewhere - for example in a sphere there are an infinite number of shortest paths between the north pole and the south pole.

In general, the distance between two points measured in this metric along such a geodesic is

$$d_{\mathbb{H}}(x_1 + y_1 i, x_2 + y_2 i) = \operatorname{arccosh}\left(1 + \frac{(x_2 - x_1)^2 + (y_2 - y_1)^2}{2y_1 y_2}\right). \qquad (1.1)$$

1.1.2 Lines and circles

To develop these first ideas further we can identify $\mathbb{R}^2$ with $\mathbb{C}$ by identifying the point $(x, y) \in \mathbb{R}^2$ with the complex number $x + iy \in \mathbb{C}$. Now we can establish expressions for lines and circles.

A straight line L in $\mathbb{R}^2$ has the form

$$ax + by + c = 0$$

for some choice of $a, b, c \in \mathbb{R}$. Writing $z = x + iy$ and recalling that the complex conjugate of z is given by $\bar{z} = x - iy$ we can see that

$$x = \frac{1}{2}(z + \bar{z}), \quad y = \frac{1}{2i}(z - \bar{z}).$$

Substituting these expressions into the equation for L in $\mathbb{R}^2$ gives

$$\begin{aligned} a\left(\frac{1}{2}(z + \bar{z})\right) + b\left(\frac{1}{2i}(z - \bar{z})\right) + c = 0 \\ \implies \frac{1}{2}(a - ib)z + \frac{1}{2}(a + ib)\bar{z} + c = 0. \end{aligned}$$

Letting $\alpha = (a - ib)/2$ means that the equation of L is now

$$\alpha z + \bar{\alpha}\bar{z} + c = 0.$$

Let C be a circle in $\mathbb{R}^2$ with centre (x_0, y_0) and radius r. Then C has the equation $(x - x_0)^2 + (y - y_0)^2 = r^2$. Let $z = x + iy$ and $z_0 = x_0 + iy_0$ and

this equation for C then becomes $|z - z_0|^2 = r^2$. Recalling that $|w|^2 = w\bar{w}$ for a complex number $w \in \mathbb{C}$, we can write this equation as

$$(z - z_o)\,(\overline{z - z_0}) = r^2.$$

Expanding this out and also recalling that $\overline{z - z_0} = \bar{z} - \bar{z_0}$ we have

$$z\bar{z} - \bar{z_0}z - z_0\bar{z_0} - r^2 = 0.$$

Let $\beta = -\bar{z_0}$ and $\gamma = z_0\bar{z_0} - r^2 = |z_0|^2 - r^2$. Then C now has the equation

$$z\bar{z} + \beta z + \bar{\beta}\bar{z} + \gamma = 0.$$

1.2 Möbius transformations

An isometry is a map that preserves distances. In $\mathbb{R}^2$ some of the examples of isometries using the Euclidean distance metric

$$d(p, q) = \sqrt{(q_1 - p_1)^2 + (q_2 + p_2)^2} \tag{1.2}$$

where $p = (p_1, p_2)$ and $q = (q_1, q_2)$ include the identity map, translations, rotations, and reflections along with the compositions of any such maps.

In hyperbolic geometry the maps which preserve distances are known as Möbius transformations, taking their name in honour of German mathematician and theoretical astronomer August Möbius (1790 - 1868).

Definition 1.2.1. Let $a, b, c, d \in \mathbb{R}$ be such that $ad - bc > 0$ and define the map $\gamma : \mathbb{H} \to \mathbb{H}$ by

$$\gamma(z) = \frac{az + b}{cz + d}. \tag{1.3}$$

Transformations of $\mathbb{H}$ of this form are called *Möbius transformations of* $\mathbb{H}$.

There are three basic types of these Möbius transformations. The first is dilations of the form $\gamma(z) = az$, the second is translations of the form $\gamma(z) = z + b$, and the third is inversion $\gamma(z) = -\frac{1}{z}$.

Remark 1.2.2. At this point it might be a good idea to check that if γ is a Möbius transformation of $\mathbb{H}$ then γ is a well-defined map $\gamma : \mathbb{H} \to \mathbb{H}$ and to show that γ maps $\mathbb{H}$ to itself bijectively.

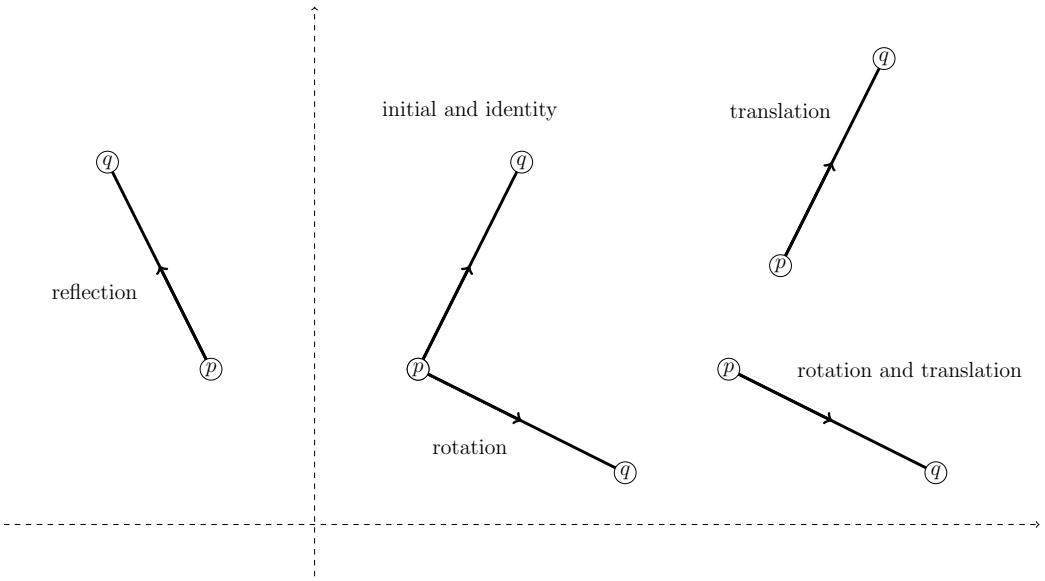


Figure 1.2: Examples of some (very) basic isometries in Euclidean geometry.

Firstly we show that γ maps $\mathbb{H}$ to itself. Let $z = x + iy \in \mathbb{H}$ and let $\gamma(z) = \gamma(z) = \frac{az+b}{cz+d}$ be a Möbius transformation of $\mathbb{H}$. Then

$$\gamma(z) = \frac{a(x+iy)+b}{c(x+iy)+d} = \frac{(ax+b+iay)(cx+d-icy)}{(cx+d+icy)(cx+d-icy)}.$$

This has imaginary part

$$\frac{1}{|cz+d|^2}(-cy(ax+b)+(cx+d)ay) = \frac{1}{|cz+d|^2}(ad-bc)y$$

which is positive. Hence γ maps $\mathbb{H}$ to itself.

Furthermore, if $\gamma(z) = (az+b)/(cz+d)$ then letting $w = (az+b)/(cz+d)$ and solving for z in terms of w shows that $\gamma^{-1}(z) = (dz-b)/(-cz+a)$. Hence γ^{-1} exists and so γ is a bijection.

Interestingly if we scale each of the variables a, b, c, and d by an identical amount we end up with the same Möbius transformation as we would have had without the scaling. This is simple enough to show when we begin with the standard writing of a Möbius transformation where each component is scaled by $k \in \mathbb{R}$,

$$\frac{kaz+kb}{kcz+kd} = \frac{k(az+b)}{k(cz+d)} = \frac{az+b}{cz+d}.$$

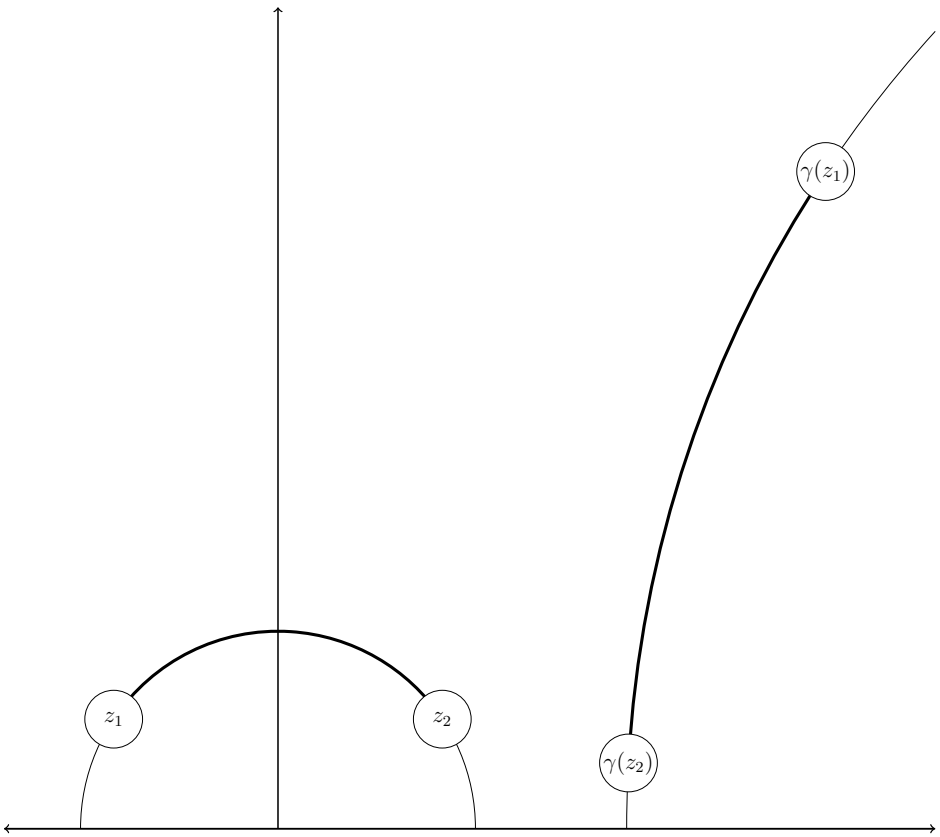


Figure 1.3: Example of a Möbius transformation of $\mathbb{H}$. Here $z_1 = -3 + 2i$, $z_2 = 3 + 2i$, and the transformation is $\gamma(z) = (10z + 6)/(z + 3)$.

Now we can work on showing that the Möbius transformations we have defined are in fact isometries.

Proposition 1.2.3. *Let γ be a Möbius transformation of $\mathbb{H}$. Let $z, z' \in \mathbb{H}$ and let σ be a path from z to z'. Then* $\text{length}_{\mathbb{H}}(\gamma \circ \sigma) = \text{length}_{\mathbb{H}}(\sigma)$.

Proof. Let $y(z) = (az + b)/(cz + d)$ where $a, b, c, d \in \mathbb{R}$ and $ad - bc > 0$. It is an easy calculation to check that for any $z \in \mathbb{H}$

$$|\gamma'(z)| = \frac{ad - bc}{|cz + d|^2}$$

and

$$\mathrm{Im}(\gamma(z)) = \frac{ad-bc}{|cz+d|^2}\mathrm{Im(z)}.$$

Let $\sigma : [0,1] \to \mathbb{H}$ be a parametrisation of σ. Then, by using the chain rule,

$$\begin{aligned}
\mathrm{length}_{\mathbb{H}}(\gamma \circ \sigma) &= \int_0^1 \frac{|(\gamma \circ \sigma)'(t)|}{\mathrm{Im}(\gamma \circ \sigma)(t)} dt \\
&= \int_0^1 \frac{|\gamma'(\sigma(t))||\sigma'(t)|}{\mathrm{Im}(\gamma \circ \sigma)(t)} dt \\
&= \int_0^1 \frac{ad-bc}{c\sigma(t)+d|^2} |\sigma'(t)| \frac{|c\sigma(t)+d|^2}{ad-bc} \frac{1}{\mathrm{Im}(\sigma(t))} dt \\
&= \int_0^1 \frac{|\sigma'(t)|}{\mathrm{Im}(\sigma(t))} dt \\
&= \mathrm{length}_{\mathbb{H}}(\sigma).
\end{aligned}$$

□

Proposition 1.2.4. *Let γ be a Möbius transformation of $\mathbb{H}$. Then γ is an isometry of $\mathbb{H}$. That is, for any $z, z' \in \mathbb{H}$ we have*

$$d_{\mathbb{H}}(\gamma(z), \gamma(z')) = d_{\mathbb{H}}(z, z').$$

Proof. We first note that if σ is a path from z to z' then $\gamma \circ \sigma$ is a path from $\gamma(z)$ to $\gamma(z')$. Moreover, any path from $\gamma(z)$ to $\gamma(z')$ arises in this way. By Proposition 1.2.3 we have that $\mathrm{length}_{\mathbb{H}}(\gamma \circ \sigma) = \mathrm{length}_{\mathbb{H}}(\sigma)$. Taking the infimum over all paths from z to z' proves that γ is an isometry of $\mathbb{H}$. □

Here is an example using the transformation displayed in Figure 1.3 to show that despite how things may look to us, the distance between two points is conserved by Möbius transformations of the upper half-plane.

Example 1.2.5 (The transformation displayed in Figure 1.3). Let $z_1 = -3+2i$ and $z_2 = 3+2i$. Using the Möbius transformation

$$\gamma(z) = \frac{10z+6}{z+3}$$

where $a = 10, b = 6, c = 1$, and $d = 3$ we obtain $\gamma(z_1) = 10 + 12i$ and $\gamma(z_2) = \frac{32}{5} + \frac{6}{5}i$. Note that $ad - bc = 24 > 0$ and so this is a valid Möbius transformation of $\mathbb{H}$. Proposition 1.2.4 tells us we should have $d_{\mathbb{H}}(z_1, z_2) = d_{\mathbb{H}}(\gamma(z_1), \gamma(z_2))$. Here is an explicit check using the formula we have for calculating hyperbolic distance, (1.1):

$$\begin{aligned} d_{\mathbb{H}}(-3+2i, 3+2i) &= \operatorname{arccosh}\left(1 + \frac{(3--3)^2 + (2-2)^2}{2 \cdot 2 \cdot 2}\right) \\ &= \operatorname{arccosh}\left(1 + \frac{36}{8}\right) = \operatorname{arccosh}\left(\frac{11}{2}\right) \\ &= 2.38953, \\ d_{\mathbb{H}}(10+12i, \frac{32}{5} + \frac{6}{5}i) &= \operatorname{arccosh}\left(1 + \frac{(\frac{32}{5} - 10)^2 + (\frac{6}{5} - 12)^2}{2 \cdot 12 \cdot \frac{6}{5}}\right) \\ &= \operatorname{arccosh}\left(1 + \frac{\frac{324}{25} + \frac{2916}{25}}{\frac{144}{5}}\right) = \operatorname{arccosh}\left(\frac{11}{2}\right) \\ &= 2.38953. \end{aligned}$$

Proposition 1.2.6. *Let* Möb($\mathbb{H}$) *denote the set of all Möbius transformations of* $\mathbb{H}$. *Then* Möb($\mathbb{H}$) *is a group under composition.*

Proof. To prove this proposition we need to show that the composition of two Möbius transformations of $\mathbb{H}$ is itself a Möbius transformation of $\mathbb{H}$, check associativity, show that the identity map is a Möbius transformation of $\mathbb{H}$, and show that if $\gamma \in$ Möb($\mathbb{H}$) is a Möbius transformation of $\mathbb{H}$ then γ^{-1} exists and is also a Möbius transformation of $\mathbb{H}$.

(i) Let $\gamma_1 = (a_1 z + b_1)/(c_1 z + d_1)$ and $\gamma_2 = (a_2 z + b_2)/(c_2 z + d_2)$. Then their composition will be

$$\begin{aligned} \gamma_2 \circ \gamma_1 = \gamma_2(\gamma_1(z)) &= \frac{a_2\left(\frac{a_1 z + b_1}{c_1 z + d_1}\right) + b_2}{c_2\left(\frac{a_1 z + b_1}{c_1 z + d_1}\right) + d_2} \\ &= \frac{(a_1 a_2 + c_1 b_2)z + (b_1 a_2 + d_1 b_2)}{(a_1 c_2 + c_1 d_2)z + (b_1 c_2 + d_1 d_2)}. \end{aligned}$$

This is a valid Möbius transformation of $\mathbb{H}$ since

$$\begin{aligned}(a_1a_2 + c_1b_2)(b_1c_2 + d_1d_2) - (b_1a_2 + d_1b_2)(a_1c_2 + c_1d_2)\\ = (a_1d_1 - b_1c_1)(a_2d_2 - b_2c_2) > 0.\end{aligned}$$

(ii) Composition of functions is always associative, so

$$(\gamma_1 \circ \gamma_2) \circ \gamma_3 = \gamma_1 \circ (\gamma_2 \circ \gamma_3).$$

(iii) Taking $a = d = 1$ and $b = c = 0$ it is plain to see that the identity map $\gamma(z) = z$ is a Möbius transformation of $\mathbb{H}$.

(iv) It follows from Remark 1.2.2 that if γ is a Möbius transformation of $\mathbb{H}$ then so is γ^{-1}.

□

One thing to note here is the similarity of the composition of Möbius transformations in part (i) of the proof and matrix multiplication. If we instead represent $\gamma_1 = (a_1z + b_1)/(c_1z + d_1)$ and $\gamma_2 = (a_2z + b_2)/(c_2z + d_2)$ as the matrices

$$\gamma_1 = \begin{pmatrix} a_1 & c_1 \\ b_1 & d_1 \end{pmatrix} \qquad \text{and} \qquad \gamma_2 = \begin{pmatrix} a_2 & c_2 \\ b_2 & d_2 \end{pmatrix}$$

then multiplying them gives

$$\gamma_1\gamma_2 = \begin{pmatrix} a_1 & c_1 \\ b_1 & d_1 \end{pmatrix} \begin{pmatrix} a_2 & c_2 \\ b_2 & d_2 \end{pmatrix} = \begin{pmatrix} a_1a_2 + c_1b_2 & a_1c_2 + c_1d_2 \\ b_1a_2 + d_1b_2 & b_1c_2 + d_1d_2 \end{pmatrix}$$

which matches up exactly with

$$\gamma_1 \circ \gamma_2 = \frac{(a_1a_2 + c_1b_2)z + (b_1a_2 + d_1b_2)}{(a_1c_2 + c_1d_2)z + (b_1c_2 + d_1d_2)}.$$

Hence composing Möbius transformations is akin to matrix multiplication and the group of Möbius transformations is isomorphic to the 2×2 projective special linear group $\mathrm{PSL}_2(\mathbb{R})$, the quotient group $\mathrm{SL}_2(\mathbb{R})/\{\pm I\}$ where $\mathrm{SL}_2(\mathbb{R})$ is the group of 2×2 real matrices with determinant one - something which will rear its head again in a later section.

Now let us look at some more of the properties of these transformations and what they can tell us, starting with a lemma that is needed in a later proof.

Lemma 1.2.7. *Let H have the form $\alpha z\overline{z} + \beta z + \beta\overline{z} + \gamma = 0$. Then there exists $\gamma \in M\ddot{o}b(\mathbb{H})$ such that γ maps H bijectively to the imaginary axis.*

Proof. If H is the vertical line $\mathrm{Re}(z) = a$ then the translation mapping z to $z - a$ is a Möbius transformation of $\mathbb{H}$ that maps H to the imaginary axis $\mathrm{Re}(z) = 0$.

Let H be a semi-circle with end-points $\eta_-, \eta_+ \in \mathbb{R}$ with $\eta_- < \eta_+$. Note that the imaginary axis is a unique element with end-points at zero and infinity. Consider the map

$$\gamma(z) = \frac{z - \eta_+}{z - \eta_-}.$$

As $-\eta_- + \eta_+ > 0$ this is a Möbius transformation of $\mathbb{H}$. Clearly $\gamma(\eta_+) = 0$ and $\gamma(\eta_-) = \infty$, so $\gamma(H)$ must be the imaginary axis. □

Definition 1.2.8. Let γ be a Möbius transformation of $\mathbb{H}$. We say that a point $z_0 \in \mathbb{H} \cup \partial\mathbb{H}$ is a *fixed point* of γ if

$$\gamma(z_0) = \frac{az_0 + b}{cz_0 + d} = z_0.$$

One easy example to give here is that every point $z \in \mathbb{H} \cup \partial\mathbb{H}$ is a fixed point of the identity Möbius transformation γ_{id} where we have $a = 1, b = 0, c = 0$, and $d = 1$ since

$$\gamma_{\mathrm{id}}(z) = \frac{z + 0}{0 + 1} = z.$$

The identity map is the only one which has this property so let us look at cases where the transformation is not the identity.

Proposition 1.2.9. *Let γ be a Möbius transformation of $\mathbb{H}$ and suppose that γ is not the identity. Then γ has either:*

(i) two fixed points in $\partial\mathbb{H}$ and none in $\mathbb{H}$;

(ii) one fixed point in $\partial\mathbb{H}$ and none in $\mathbb{H}$;

(iii) no fixed points in $\partial\mathbb{H}$ and one in $\mathbb{H}$.

Corollary 1.2.10. *Suppose γ is a Möbius transformation of $\mathbb{H}$ with three or more fixed points. Then γ is the identity - and so fixes every point.*

Each of the three types of Möbius transformations described in the cases above has its own name.

Definition 1.2.11. Let γ be a Möbius transformation of $\mathbb{H}$. We will say that

(i) γ is *hyperbolic* if it has two fixed points in $\partial\mathbb{H}$ and none in $\mathbb{H}$;

(ii) γ is *parabolic* if it has one fixed point in $\partial\mathbb{H}$ and none in $\mathbb{H}$;

(iii) γ is *elliptic* if it has no fixed points in $\partial\mathbb{H}$ and one in $\mathbb{H}$.

1.3 Geodesics in the upper half-plane

As briefly mentioned before, the infimum of all paths between two points is known as a geodesic. When using the Euclidean distance metric (1.2) in $\mathbb{R}^2$ the geodesics are the straight lines between any two given points. Things are different in hyperbolic geometry and they also vary depending on which model is being used to work with.

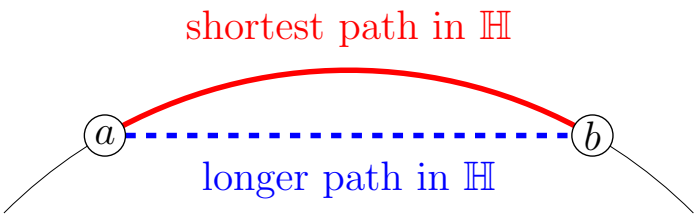


Figure 1.4: An explicit illustration of how geodesics being shortest paths in hyperbolic space might not be immediately compatible with a brain that has only known Euclidean geometry.

Here is a proposition which uses Möbius transformations discussed in the previous section and is needed to prove that the geodesics in $\mathbb{H}$ are the semi-circles orthogonal to the real axis and the vertical straight lines as well as their uniqueness.

Proposition 1.3.1. *Let H be either (i) a semi-circle orthogonal to the real axis, or (ii) a vertical straight line. Let γ be a Möbius transformation of $\mathbb{H}$. Then $\gamma(H)$ is either a semi-circle orthogonal to the real axis or a vertical straight line.*

Proof. We have already shown that the Möbius transformations of $\mathbb{H}$ map the upper half-plane to itself bijectively and so it is sufficient to show that γ maps vertical straight lines in $\mathbb{C}$ and circles in $\mathbb{C}$ with real centres to vertical straight lines and circles with real centres.

In $\mathbb{C}$ these are given by an equation of the form

$$\alpha z\overline{z} + \beta z + \beta\overline{z} + \gamma = 0 \tag{1.4}$$

for some $\alpha, \beta, \gamma \in \mathbb{R}$. Letting

$$w = \gamma(z) = \frac{az+b}{cz+d}$$

implies that

$$z = \frac{dw-b}{-cw+a}.$$

Substituting this into (1.4) we acquire

$$\alpha\left(\frac{dw-b}{-cw+a}\right)\left(\frac{d\overline{w}-b}{-c\overline{w}+a}\right) + \beta\left(\frac{dw-b}{-cw+a}\right) + \beta\left(\frac{d\overline{w}-b}{-c\overline{w}+a}\right) + \gamma = 0.$$

Multiplying by $(-cw+a)(-c\overline{w})$ gives

$$\begin{aligned}\alpha(dw-b)(d\overline{w}-b) &+ \beta(dw-b)(-c\overline{w}+a)\\ &+ \beta(d\overline{w}-b)(-cw+a) + \gamma(-cw+a)(-c\overline{w}+a) = 0.\end{aligned}$$

From here we can expand and then gather terms together to obtain

$$\begin{aligned}(\alpha d^2 - 2\beta cd + \gamma c^2)w\overline{w} &+ (-\alpha bd + \beta ad + \beta bc - \gamma ac)w\\ &+ (-\alpha bd + \beta ad + \beta bc - \gamma ac)\overline{w} + (\alpha b^2 - 2\beta ab + \gamma a^2) = 0.\end{aligned} \tag{1.5}$$

When we let

$$\begin{aligned}\alpha' &= \alpha d^2 - 2\beta cd + \gamma c^2,\\ \beta' &= -\alpha bd + \beta ad + \beta bc - \gamma ac,\\ \gamma' &= \alpha b^2 - 2\beta ab + \gamma a^2,\end{aligned}$$

we see that (1.5) has the form $\alpha' z\overline{z} + \beta' z + \beta'\overline{z} + \gamma' = 0$ with $\alpha', \beta', \gamma' \in \mathbb{R}$ and this is the equation of either a vertical line or a circle with real centre as defined by (1.4). $\square$

Now we have all the requisite tools to write out a formal theorem and proof describing the geodesics in the upper half-plane.

Theorem 1.3.2. *The geodesics in $\mathbb{H}$ are the semi-circles orthogonal to the real axis and the vertical straight lines. Moreover, given any two points in $\mathbb{H}$ there exists a unique geodesic passing through them.*

Proof. Let $z, z' \in \mathbb{H}$. Then we can always find a unique element H with the form in (1.4) containing z, z'. If z and z' have the same real part then H will be a vertical straight line, otherwise H will be a semi-circle with a real centre. Let σ be any path from z to z'.

Apply the Möbius transformation $\gamma \in \text{Möb}(\mathbb{H})$ constructed in Lemma 1.2.7 so that $\gamma(z), \gamma(z')$ lie on the imaginary axis. Then $\gamma \circ \sigma$ is a path from $\gamma(z)$ to $\gamma(z')$. By Proposition 1.2.3 we have that $\text{length}_{\mathbb{H}}(\sigma) = \text{length}_{\mathbb{H}}(\gamma \circ \sigma)$.

Let $a \leq b$. We will then assume that the hyperbolic distance between ia and ib is $\log(b/a)$. Moreover, the vertical line joining ia to ib is the unique path between ia and ib with length $\log(b/a)$ and any other path from ia to ib has length strictly greater than $\log(b/a)$.

Using this the imaginary axis is the unique geodesic passing through $\gamma(z)$ and $\gamma(z')$. Hence $\text{length}_{\mathbb{H}}(\gamma \circ \sigma)$ achieves its infimum precisely when $\gamma \circ \sigma$ is the arc of imaginary axis from $\gamma(z)$ to $\gamma(z')$.

Hence $\text{length}_{\mathbb{H}}(\sigma)$ achieves its infimum when $\gamma \circ \sigma$ is the imaginary axis from $\gamma(z)$ to $\gamma(z')$, i.e., when σ is the image under σ^{-1} of the imaginary axis from $\gamma(z)$ to $\gamma(z')$. As $\gamma^{-1} \in \text{Möb}(\mathbb{H})$, it follows from Proposition 1.3.1 that σ is an arc of straight line or semi-circle with real centre passing through z and z'. □

The vertical straight lines match up with what we have in Euclidean geometry but the semi-circles are an obvious difference. In an alternate hyperbolic universe sports such as golf and baseball might look a lot different and be more difficult for earthlings living in this universe to excel at [9].

1.4 The Poincaré disc model

The upper half-plane is perhaps the simplest model of hyperbolic geometry to begin with because of how all of its points can be identified with complex

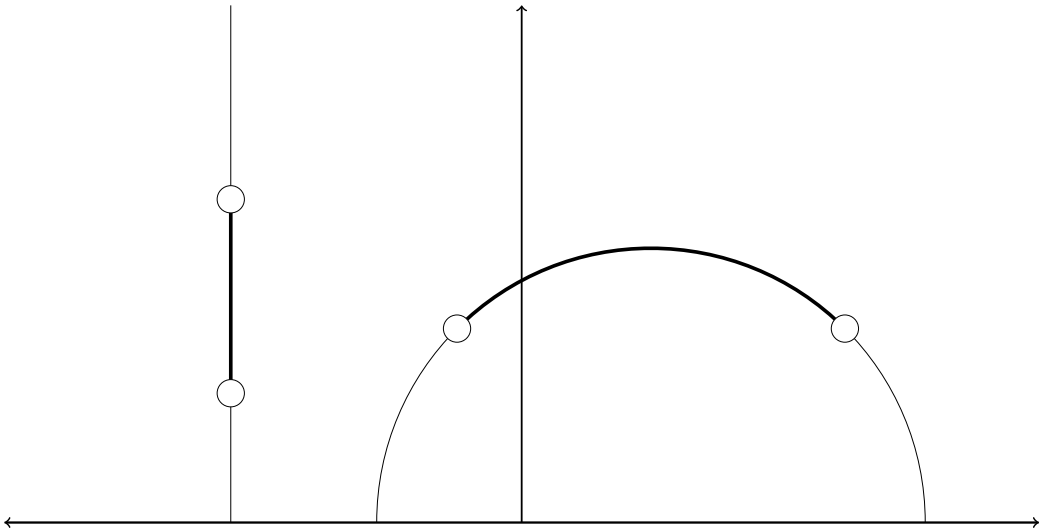


Figure 1.5: Examples of geodesics between points in the upper half-plane.

numbers in a straightforward fashion. The other model that will be looked at in this paper is the Poincaré disc model, named after French polymath Henri Poincaré (1854 - 1912). One big advantage of this model is that it is contained entirely within the unit circle which can be particularly useful when we want to create visuals.

Definition 1.4.1. The disc $\mathbb{D} = \{z \in \mathbb{C} : |z| < 1\}$ is called the *Poincaré disc*. The circle $\partial\mathbb{D} = \{z \in \mathbb{C} : |z| = 1\}$ is called the *circle at* ∞ or *boundary* of $\mathbb{D}$.

When this is equipped with the metric

$$ds^2 = \frac{dx^2 + dy^2}{(1 - x^2 - y^2)^2}$$

we have our second model of the hyperbolic plane.

The geodesics in the Poincaré disc model of hyperbolic geometry are the arcs of circles and diameters in $\mathbb{D}$ that meet $\partial\mathbb{D}$ orthogonally. Rather than developing an analysis analogous to that of the upper half-plane it is quicker and more convenient to transfer results from the upper half-plane directly to the Poincaré disc.

To do this we can consider the map $h : \mathbb{H} \to \mathbb{D}$ defined by

$$h(z) = \frac{z - i}{iz - 1}. \tag{1.6}$$

There is also an inverse map $g : \mathbb{D} \to \mathbb{H}$ allowing us to go from the Poincaré disc back to the upper half-plane which is defined by

$$g(z) = \frac{-z + i}{-iz + 1}. \tag{1.7}$$

1.5 Geodesics in the Poincaré disc

Not only do geodesics in hyperbolic geometry differ to those in Euclidean geometry but they also differ as we move between the many different models of hyperbolic geometry. The geodesics in the Poincaré disc take a different form but can perhaps most easily be thought of as the images under h from (1.6) of the geodesics in the upper half-plane.

Proposition 1.5.1. *The geodesics in the Poincaré disc are the diameters of $\mathbb{D}$ and the arcs of circles in $\mathbb{D}$ that meet $\partial\mathbb{D}$ at right-angles.*

Outline of a proof. It can be shown that the map $h : \mathbb{H} \to \mathbb{D}$ (1.6) is a conformal map, that is, it preserves angles. Then using the expressions of lines and circles in $\mathbb{C}$ as solutions to $\alpha z\overline{z} + \beta z + \beta\overline{z} + \gamma = 0$ it is possible to show that h maps the set of all circles and lines in $\mathbb{C}$ to itself. Since h maps $\partial\mathbb{H}$ to its analgoue $\partial\mathbb{D}$ and the geodesics in the upper half-plane are the arcs of circles and lines that meet $\partial\mathbb{H}$ orthogonally, the conformality of h means that the image in the Poincaré disc of a geodesic in the upper half-plane is a circle or line that meets $\partial\mathbb{D}$ orthogonally. □

Remark 1.5.2 (Historical aside which gives some context as to what inspired this paper). The work of art by M. C. Escher featured in Figure 1.1 at the beginning of this chapter is a tessellation of hyperbolic space presented in the Poincaré disc. By his own admission Escher was not mathematically trained and his understanding of mathematics was largely visual and intuitive. A publication by Canadian mathematician H. S. M. Coxeter described how a tessellation from the centre to the edge of a circle is increasingly reduced and the motifs come to lie infinitely close together. In 1957 Coxeter gave a talk

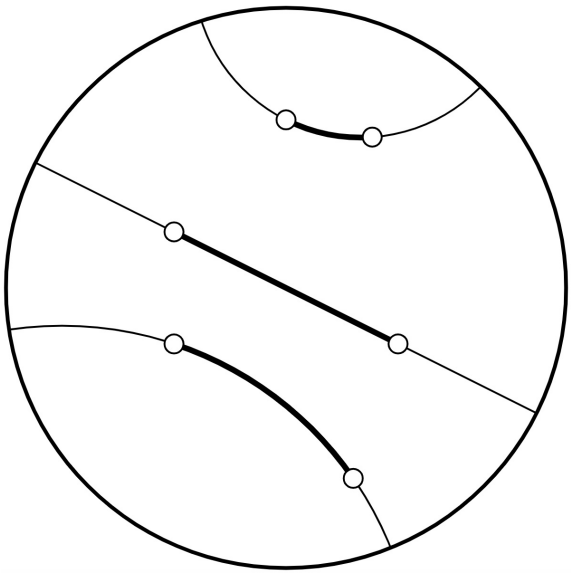


Figure 1.6: Examples of geodesics between points in the Poincaré disc.

for the Royal Society of Canada and he asked Escher by letter if he could use a few of the graphic artist's works in the lecture. Afterwards Coxeter sent Escher a copy of his lecture (which had been published under the name Crystal Symmetry and its Generalizations) in which he also included a figure with which Escher would become enamored.

After mastering the hyperbolic tilings and producing three previous works (Circle Limits I, II, and III) Escher made a fourth and final piece. This one was based on a drawing from one of his sketchbooks: 'Regular Division drawing no. 45', done during Christmas 1941 [10]. This is shown in Figure 1.7 along with Circle Limit IV again.

1.6 Other models

There are other models available each with their own advantages and disadvantages. Some of these include the Klein model, the hyperboloid model, the hemisphere model, the Gans model, and the band model. All of the models describe essentially the same structure but the difference between them is that they represent different coordinate charts laid down on the same metric

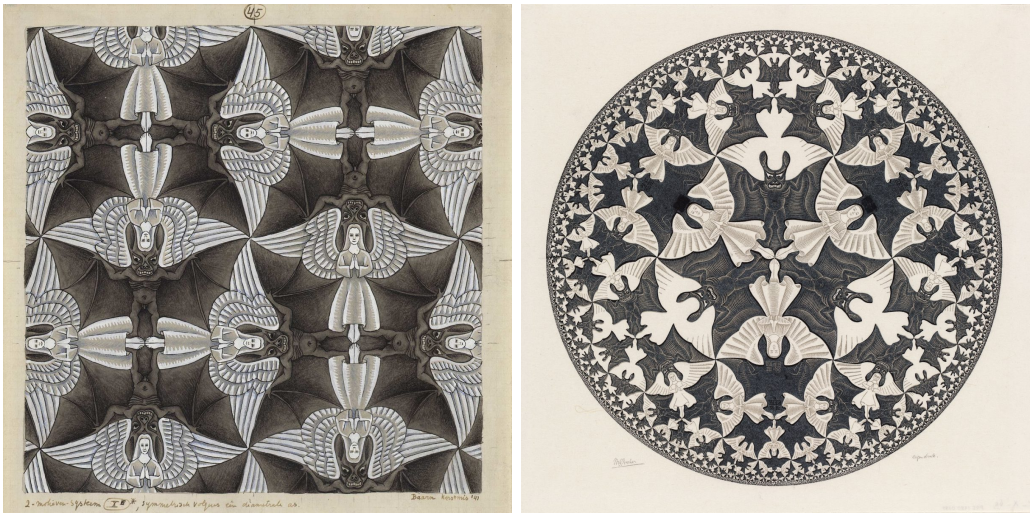


Figure 1.7: *Left*: Regular Division drawing no. 45, 1941. *Right*: Circle Limit IV (Heaven and Hell), 1960. Both by M. C. Escher.

space, the hyperbolic plane.

The four most commonly used models are the upper half-plane, the Poincaré disc, the Klein, and the hyperboloid however the only two which will be worked with any further are the first two in that list.

1.7 Fuchsian groups

Fuchsian groups are a basic example of lattices in semi-simple Lie groups. Their very solid nature has allowed people to illustrate their many features having far-reaching generalisations in geometry and number theory. Some such areas include differential geometry, number theory, Lie theory, and representation theory.

Think back to the discussion of the Möbius transformations of $\mathbb{H}$ and recall that together they form a group under composition. The same is true of the Möbius transformations of $\mathbb{D}$. There are many subgroups of these groups and one class of subgroups in particular will be most relevant to us here.

Definition 1.7.1. A *Fuchsian group* is a discrete subgroup of either Möb($\mathbb{H}$), the Möbius transformations of $\mathbb{H}$, or Möb($\mathbb{D}$), the Möbius transformations of

$\mathbb{D}$.

General Fuchsian groups were first studied in an 1882 paper by Henri Poincaré [11] who was motivated by the 1880 paper [12] and therefore named them after that paper's author, the German mathematician Lazarus Fuchs (1833 - 1902) and perhaps the most important Fuchsian group is $\mathrm{PSL}_2(\mathbb{R})$ since it is isomorphic to the group of all orientation-preserving isometries of $\mathbb{H}$.

Here is an alternative definition of Fuchsian group from [13], where $\mathrm{Isom}(\mathbb{H})$ is the group of isometries of the upper half-plane:

Definition 1.7.2. A discrete subgroup of $\mathrm{Isom}(\mathbb{H})$ is called a *Fuchsian group* if it consists of orientation-preserving transformations, in other words, a Fuchsian group is a discrete subgroup of $\mathrm{PSL}_2(\mathbb{R})$.

Comparing these two definitions makes it clear that $\mathrm{Möb}(\mathbb{H}) \cong \mathrm{PSL}_2(\mathbb{R})$.

1.8 Hyperbolic polygons

In Euclidean geometry, an n-sided polygon is a subset of the Euclidean plane bounded by n straight lines. Thus the edges of a Euclidean polygon are formed by segments of Euclidean geodesics. A hyperbolic polygon is defined in a corresponding manner.

Definition 1.8.1. Let $z, w \in \mathbb{H} \cup \partial\mathbb{H}$. Then there exists a unique geodesic that passes through both z and w. We denote by $[z, w]$ the part of this geodesic that connects z and w. We call $[z, w]$ the *segment* or *arc* of geodesic between z and w.

Definition 1.8.2. Let $z_1, \ldots z_n \in \mathbb{H} \cup \partial\mathbb{H}$. Then the *hyperbolic n-gon P* with vertices at $z_1, \ldots, z_n$ is the region of $\mathbb{H}$ bounded by the geodesic segments

$$[z_1, z_2], [z_2, z_3], \ldots, [z_{n-1}, z_n], [z_n, z_1].$$

Notice that we allow some of the vertices to lie on the boundary of the hyperbolic plane. Such a vertex is called an ideal vertex. If all the vertices of a polygon P lie on $\partial\mathbb{H}$ then we call P an ideal polygon. Notice that the angle at an ideal vertex is zero - this is because all geodesics meet $\partial\mathbb{H}$ at right-angles and so the angle between any two such geodesics is zero.

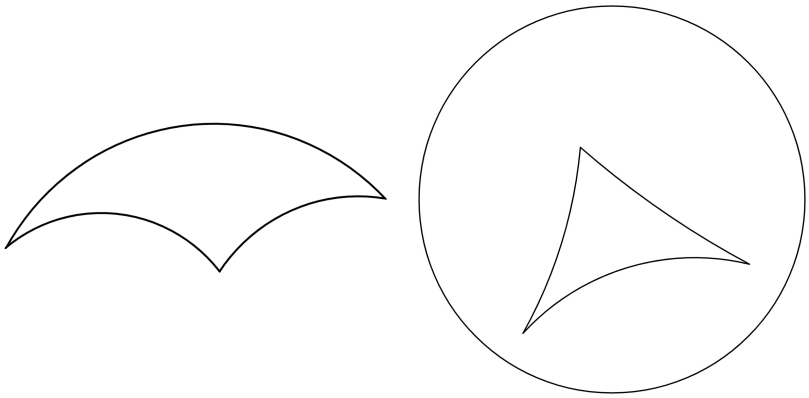


Figure 1.8: Examples of a hyperbolic triangle: *left* in the upper half-plane model, *right* in the Poincaré disc.

It is the ideal polygons and their tessellations that will come in handy as we reach the end of the next chapter. To finish this chapter we will state two more theorems, one without proof and the other with a partial proof. The final theorem enables us to be able to tell if certain shapes will entirely tile hyperbolic space which again becomes useful towards the next chapter's end.

Theorem 1.8.3 (Gauss-Bonnet Theorem for a hyperbolic polygon)**.** *Let P be an n-sided hyperbolic polygon with vertices $v_1, \ldots, v_n$ and internal angles $\alpha_1, \ldots, \alpha_n$. Then*

$$\text{Area}_{\mathbb{H}}(P) = (n-2)\pi - (\alpha_1 + \cdots + \alpha_n). \tag{1.8}$$

Theorem 1.8.4. *There exists a tessellation of the hyperbolic plane by regular hyperbolic n-gons with k of these n-gons meeting at each vertex if and only if*

$$\frac{1}{n} + \frac{1}{k} < \frac{1}{2}. \tag{1.9}$$

Partial proof. We only prove that if there is a tessellation then n and k satisfy (1.9), the converse is much harder. Let α denote the internal angle of a regular n-gon P. Then as k such polygons meet at each vertex we must have that $\alpha = 2\pi/k$. As the area of the polygon P must be positive, substituting $\alpha = 2\pi/k$ into (1.8) and rearranging we have

$$\frac{1}{n} + \frac{1}{k} < \frac{1}{2}.$$

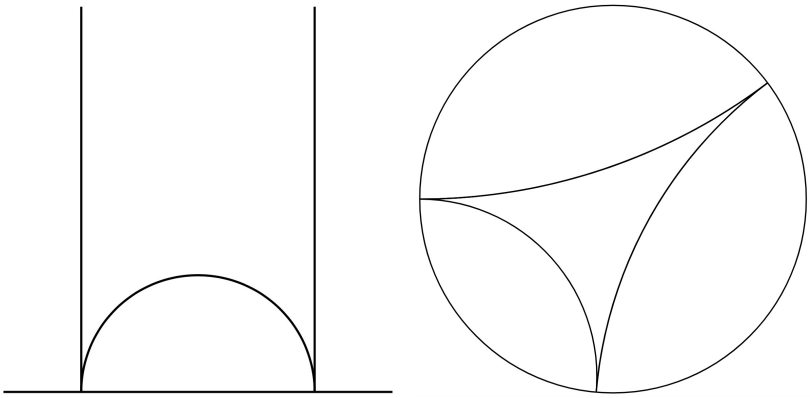


Figure 1.9: Examples of an ideal hyperbolic triangle: *left* in the upper half-plane model, *right* in the Poincaré disc.

□

Chapter 2

Sandpiles and tilings

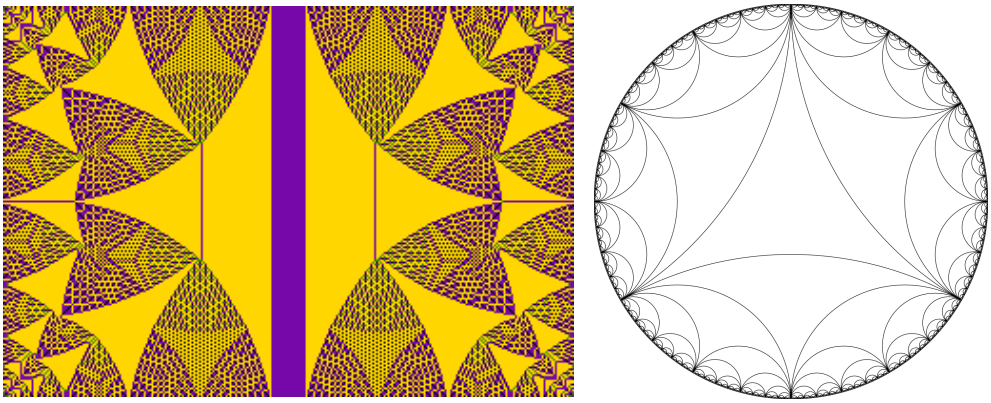


Figure 2.1: *Left*: The identity element of the sandpile group of a rectangular grid. Yellow pixels correspond to vertices carrying three particles, lilac to two particles, green to one, and black to zero. *Right*: The Poincaré disc model of the hyperbolic plane tiled with ideal triangles. *Both taken from Wikipedia.*

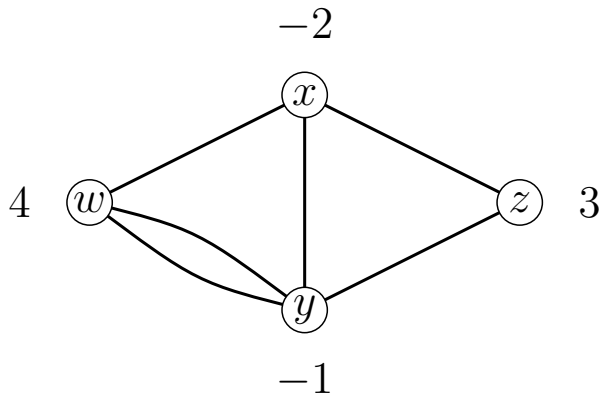
Shown in the figure above is an example of an identity element of the sandpile group of a rectangular grid alongside an ideal tiling of the hyperbolic space. As mentioned in the introduction, at the beginning of this project I set myself the ambitious goal of seeing how such a system might look when connected to a tiling of the hyperbolic space and to potentially produce some of the fractal-type images that can be seen in the more familiar grids.

2.1 Chip-firing

To work with sandpiles we need to first introduce the idea of chip-firing on graphs. One way of doing this is to head straight into an example and gain some intuition for how things work before stating some of the definitions in order to make the notation used in them more penetrable.

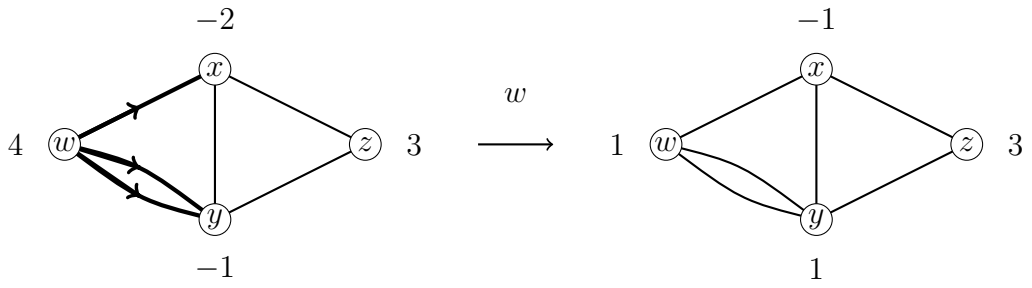
2.1.1 An initial game

Consider the graph pictured below. We think of the vertices V as individuals, the edges E as relationships between individuals, and the entire graph $G = (V, E)$ as a community. The number of edges between two vertices indicates the strength of the relationship between the corresponding individuals with zero edges indicating no relationship at all.



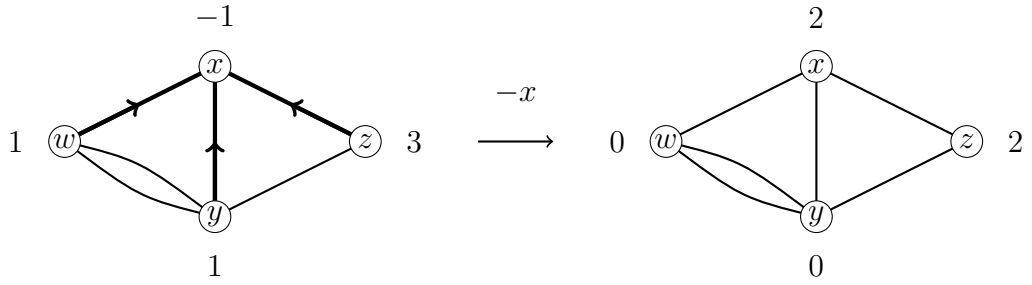
We place an integer representing a number of dollars next to each vertex, interpreting negative values as debt. Such a distribution of dollars can be represented as a formal sum of vertices. For example the distribution of wealth D in the above diagram is given by $D = 4w - 2x - y + 3z$.

To redistribute wealth each individual may choose to lend a dollar along each edge it is incident to (possibly going into debt) or else borrow a dollar along each edge (possibly forcing someone else into debt). The below diagram shows a lending move by the vertex w:



The goal is to find a sequence of lending/borrowing moves so that everyone is debt-free. This is called the dollar game on G starting from D and if such a sequence exists the game is said to be winnable.

In our example: after w has made the lending move only x is in debt. If x now borrows, then, as shown below, the dollar game is won.



2.1.2 Formal definitions

Definition 2.1.1. A *multigraph* $G = (V, E)$ is a pair consisting of a set of *vertices* V and a multiset of *edges* E comprised of unordered pairs $\{v, w\}$ of vertices. The prefix multi here means that a pair $\{v, w\}$ may occur multiple times in E. For ease of notation, we generally write vw for the edge $\{v, w\}$. A multigraph G is *finite* if both V and E are finite, and it is *connected* if any two vertices are joined by a sequence of edges.

Definition 2.1.2. A *divisor* on a graph G is an element of the free abelian group on the vertices:

$$\mathrm{Div}(G) = \mathbb{Z}V = \left\{ \sum_{v \in V} D(v)v : D(v) \in \mathbb{Z} \right\}.$$

Definition 2.1.3. The *degree* of $D = \sum_{v \in V} D(v)v \in \mathrm{Div}(G)$ is the integer

$$\deg(D) = \sum_{v \in V} D(v).$$

Definition 2.1.4. Let $D, D' \in \mathrm{Div}(G)$ and $v \in V$. Then D' is obtained from $D \in \mathrm{Div}(G)$ by a *lending* or *firing move at* v, denoted $D \xrightarrow{v} D'$, if

$$D' = D - \sum_{vw \in E} (v - w) = D - \deg_G(v)v + \sum_{vw \in E} w.$$

Similarly, D' is obtained from D by a *borrowing move at* v, denoted $D \xrightarrow{-v} D'$, if

$$D' = D + \sum_{vw \in E} (v - w) = D + \deg_G(v)v - \sum_{vw \in E} w.$$

A collection of lending/firing/borrowing moves is known as a *lending/firing/borrowing script*.

Definition 2.1.5. Let $D, D' \in \mathrm{Div}(G)$ and suppose that D' is obtained from D by lending from all of the vertices in $W \subseteq V$. Then D' is the result of a *set-lending* or *set-firing* by W, denoted $D \xrightarrow{W} D'$.

Definition 2.1.6. Let $D, D' \in \mathrm{Div}(G)$. Then D is *linearly equivalent* to D' if D' may be obtained from D by a sequence of lending moves. In that case, we write $D \sim D'$.

Definition 2.1.7. The *divisor class* determined by $D \in \mathrm{Div}(G)$ is

$$[D] = \{D' \in \mathrm{Div}(G) : D' \sim D\}.$$

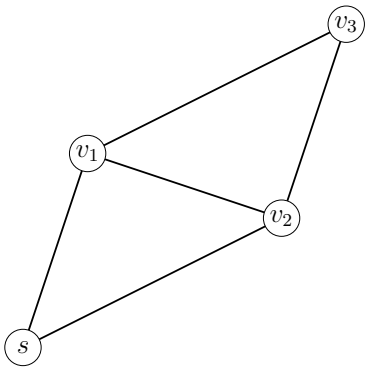
Definition 2.1.8. A divisor D on G is *effective* if $D \geq 0$ and the divisor $D \in \mathrm{Div}(G)$ is *winnable* if D is linearly equivalent to an effective divisor. Otherwise D is *unwinnable*.

2.2 Sandpiles

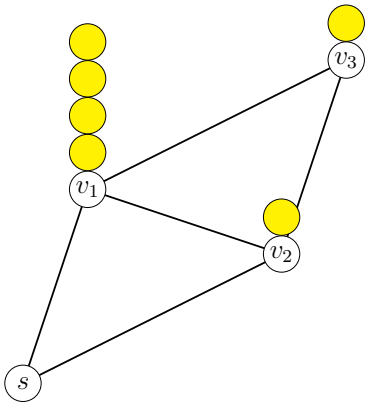
Not everything in the previous section will be needed as we move forward but it is important to understand how firing moves work on these kinds of graphs. Rather than the dollar game we will be interested in a dynamical system known as the sandpile model. In the same manner as the chip-firing section one approach of growing into the topic is to begin with an example.

2.2.1 A first example

Let G be the diamond graph pictured below:



Imagine that grains of sand - these can also be thought of as chips and called so in different scenarios - can be piled onto the vertices of G to form a sandpile. The figure below shows G with four grains of sand on v_1, one grain on each of v_2 and v_3, and no sand on s:

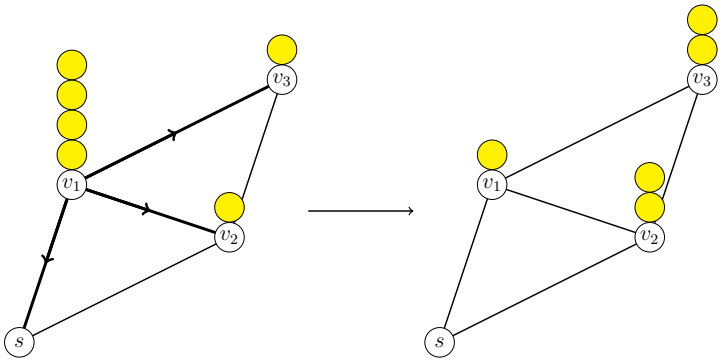


We allow only non-negative amounts of sand at each vertex. If a vertex gets too much sand it becomes unstable and at this point it may fire or topple, sending grains of sand to its neighbouring vertices. The other vertices may also be unstable or may become unstable due to the toppling of its neighbours. Thus dropping a single grain of sand onto an existing sandpile can lead to an avalanche of firings.

We will use the following rule: a vertex is unstable if it has at least as many grains of sand as its degree. Thus, a vertex v is unstable if it has enough sand

to send a grain along each edge to its neighbours without turning negative.

If sand were conserved then configurations with too many grains of sand would never stabilise - no matter how vertices were fired there would always be at least one unstable vertex. To prevent this we designate s to be the sink vertex and stipulate that any grains of sand landing on s immediately disappear. The diagram below shows what happens to the configuration $(4, 1, 1)$ on G when vertex v_1 fires:



Whenever a vertex adjacent to the sink fires a grain of sand is lost which tends to make the whole sandpile system more stable. Every initial sandpile on G will, through repeated firings, eventually stabilise. Furthermore the choice of ordering in which to fire unstable vertices has no effect on the eventual stabilisation of a sandpile. In light of this every sandpile c on G eventually reaches a unique stable state. We denote the stabilisation of c by c° and write

$$c \rightsquigarrow c^\circ.$$

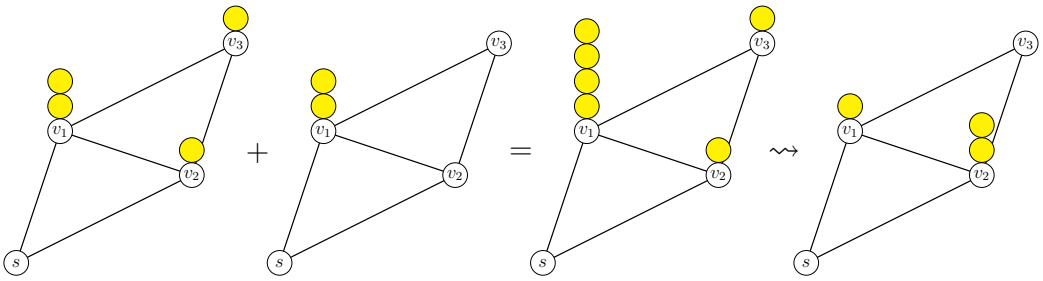
2.2.2 Additive structure

We can identify each sandpile on G with a three-tuple $c = (c_1, c_2, c_3)$ of integers by letting c_i be the number of grains of sand on vertex v_i. Let $\mathrm{Stab}(G)$ denote the collection of 18 stable sandpiles on G.

Definition 2.2.1. The *stable addition* of sandpiles a and b, denoted $a \circledast b$, is defined as ordinary addition of group elements, i.e., adding them as integer vectors, followed by stabilisation:

$$a \circledast b := (a + b)^\circ.$$

For example $(2,1,1) \circledast (2,0,0) = (1,2,0)$ as illustrated below.



$(2,1,1) \quad + \quad (2,0,0) \quad = \quad (4,1,1) \quad \rightsquigarrow \quad (1,2,0)$

We can now argue that the operation $\circledast$ is commutative and associative and that the zero sandpile - 0 = (0,0,0) - serves as an identity element.

If we let $a, b, c \in \mathrm{Stab}(G)$ then

$$a \circledast b = (a+b)^\circ = (b+a)^\circ = b \circledast a$$

and so the operation is commutative.

Since $c \geq 0$ the sequence that stabilises $a+b$ is still legal for $a+b+c$, so

$$a+b+c \rightsquigarrow (a+b)^\circ + c \rightsquigarrow ((a+b)^\circ + c)^\circ) = (a \circledast b) \circledast c.$$

Similarly the sequence that stabilises $b+c$ is legal for $a+b+c$, and we have

$$a+b+c \rightsquigarrow a + (b+c)^\circ \rightsquigarrow (a + (b+c)^\circ)^\circ = a \circledast (b \circledast c).$$

Therefore

$$(a \circledast b) \circledast c = ((a+b)^\circ + c)^\circ = (a+b+c)^\circ = (a + (b+c)^\circ)^\circ = a \circledast (b \circledast c)$$

and the operation is associative.

As for the zero sandpile being the identity element we have

$$a \circledast 0 = (a+0)^\circ = a^\circ = a, \qquad 0 \circledast a = (0+a)^\circ = a^\circ = a$$

as required.

By the above $(\mathrm{Stab}(G), \circledast)$ is almost an abelian group - it only lacks inverses. Such a structure is called a commutative monoid. It turns out that there is a special subset of $\mathrm{Stab}(G)$ that does form a group, which we now describe.

Whenever we have an unstable sandpile c it topples (perhaps producing an avalanche) to yield a stable sandpile c°. This model is dissipative because sand is lost into the sink s and it is slowly-driven because of the ongoing addition of sand at a rate that allows for stabilisation between each additional grain. Such systems often exhibit the complex behaviour of self-organised criticality.

It is this concept of self-organised criticality - first proffered by Per Bak, Chao Tang, and Kurt Wiesenfeld in a 1987 paper [14] - that governs the way the grains of sand finish up in the fractal patterns discussed at the very beginning of this chapter. Sandpile models such as these were in fact the first discovered example of a dynamical system displaying self-organised criticality.

These ideas are implemented in the following experiment:

- pick a vertex of G at random,
- drop a grain of sand on that vertex and stabilise the resulting sandpile,
- repeat until 100 grains have been dropped.

The results from ten trials of this experiment show that there are eight sandpiles that appear more than once in each trial. These are the recurrent sandpiles for G, denoted $\mathcal{S}(G)$:

$$(0,2,1), (1,2,0), (1,2,1), (2,0,1), (2,1,0), (2,1,1), (2,2,0), (2,2,1).$$

They form a group under the operation inherited from $\mathrm{Stab}(G)$. Note that $\mathcal{S}(G)$ does not contain the zero sandpile - $\mathcal{S}(G)$ has an identity, but it is not the identity of $\mathrm{Stab}(G)$. In other words, $\mathcal{S}(G)$ is not a submonoid of $\mathrm{Stab}(G)$ even though they share the same operation.

Example 2.2.2. One way of finding the identity element of $\mathcal{S}(G)$ is to create the addition table for $\mathcal{S}(G)$ and to then inspect that table.

Here is the working for the first few entries:

$$\begin{aligned}
(0,2,1)+(0,2,1) &= (0,4,2) \to (1,5,0) \to (2,2,1),\\
(0,2,1)+(1,2,0) &= (1,4,1) \to (2,1,2) \to (3,2,0)\\
&\quad \to (0,3,1) \to (1,0,2) \to (2,1,0),\\
(0,2,1)+(1,2,1) &= (1,4,2) \to (2,5,0) \to (3,2,1)\\
&\quad \to (0,3,2) \to (1,0,3) \to (2,1,1),\\
(0,2,1)+(2,0,1) &= (2,2,2) \to (3,0,3) \to (0,4,1)\\
&\quad \to (1,1,2) \to (2,2,0),\\
(0,2,1)+(2,1,0) &= (2,3,1) \to (3,0,2) \to (4,1,0)\\
&\quad \to (1,2,1).
\end{aligned}$$

Here is the full addition table:

	$(0,2,1)$	$(1,2,0)$	$(1,2,1)$	$(2,0,1)$	$(2,1,0)$	$(2,1,1)$	$(2,2,0)$	$(2,2,1)$
$(0,2,1)$	$(2,2,1)$	$(2,1,0)$	$(2,1,1)$	$(2,2,0)$	$(1,2,1)$	$(1,2,0)$	$(0,2,1)$	$(1,2,0)$
$(1,2,0)$	$(2,1,0)$	$(0,2,1)$	$(0,2,1)$	$(2,1,1)$	$(2,2,0)$	$(2,2,1)$	$(1,2,0)$	$(1,2,1)$
$(1,2,1)$	$(2,1,1)$	$(0,2,1)$	$(2,0,1)$	$(2,1,0)$	$(2,2,1)$	$(2,2,0)$	$(1,2,1)$	$(1,2,0)$
$(2,0,1)$	$(2,2,0)$	$(2,1,1)$	$(2,1,0)$	$(2,2,1)$	$(1,2,0)$	$(1,2,1)$	$(2,0,1)$	$(0,2,1)$
$(2,1,0)$	$(1,2,1)$	$(2,2,0)$	$(2,2,1)$	$(1,2,0)$	$(0,2,1)$	$(2,0,1)$	$(2,1,0)$	$(2,1,1)$
$(2,1,1)$	$(1,2,0)$	$(2,2,1)$	$(2,2,0)$	$(1,2,1)$	$(2,0,1)$	$(0,2,1)$	$(2,1,1)$	$(2,1,0)$
$(2,2,0)$	$(0,2,1)$	$(1,2,0)$	$(1,2,1)$	$(2,0,1)$	$(2,1,0)$	$(2,1,1)$	$(2,2,0)$	$(2,2,1)$
$(2,2,1)$	$(1,2,0)$	$(1,2,1)$	$(1,2,0)$	$(0,2,1)$	$(2,1,1)$	$(2,1,0)$	$(2,2,1)$	$(2,2,0)$

We can see from this table that the identity element is $(2,2,0)$.

2.2.3 Sandpile graphs

A sandpile graph is a triple $G = (V, E, s)$, where (V, E) is a graph, also referred to as G, and s is the sink. For a sandpile graph we require the designated sink vertex to be globally accessible - that is, there is some directed path from each vertex to s.

We let $\tilde{V}$ denote the non-sink vertices:

$$\tilde{V} := V \setminus \{s\}.$$

A configuration of sand on G is an element of the free abelian group on its

non-sink vertices:

$$\mathrm{Config}(G,s) := \mathrm{Config}(G) := \mathbb{Z}\tilde{V} := \left\{ \sum_{v\in\tilde{V}} c(v)v : c(v) \in \mathbb{Z} \text{ for all } v \right\}.$$

Here is a theorem, definition, and lemma which we need to construct some language allowing us prove that every configuration of sand on a sandpile graph has a stabilisation.

Theorem 2.2.3 (Least action principle)**.** *Let $c \in \mathrm{Config}(G)$, and suppose $\sigma, \tau \geq 0$ are firing scripts such that σ arises from a legal firing sequence for c and $c \xrightarrow{\tau} \tilde{c}$ with $\tilde{c}$ stable. Then $\sigma \leq \tau$.*

Definition 2.2.4. Let $u_1, \cdots, u_n$ be an ordering of the non-sink vertices of G such that $i < j$ if $d(u_i, s) < d(u_j, s)$, i.e., if u_i is closer to the sink. The *sandpile ordering* of $\mathrm{Config}(G) = \mathbb{Z}\tilde{V}$ is the total ordering $\prec$ defined as follows. Given distinct configurations a, b, let $c := a - b = \sum_{i=1}^{n} c_i u_i$. Then $a \prec b$ if

1. $\deg(a) < \deg(b)$, or
2. $\deg(a) = \deg(b)$ and $c_k > 0$ for the smallest k such that $c_k \neq 0$.

Lemma 2.2.5. *Let $\prec$ be a sandpile ordering, and let $c, \tilde{c} \in \mathbb{Z}\tilde{V}$. If $c \to \tilde{c}$ via a sequence of vertex firings, then $\tilde{c} \preceq c$.*

Proof. We may assume that $c \xrightarrow{v} \tilde{c}$ for some vertex v. If $(v, s) \in E$, then when v fires and some sand is lost to the sink. So in that case $\tilde{c} \prec c$ since $\deg(\tilde{c}) < \deg(c)$. Otherwise, since the sink is globally accessible, there exists $(v, u) \in E$ for some vertex u closer to the sink than v, i.e., $u \prec v$. When v fires, the vertex u receives sand and no vertex besides v loses sand. Hence $\tilde{c} \prec c$ since $\tilde{c}$ has more sand closer to the sink. □

Theorem 2.2.6. *Every configuration has a stabilisation.*

Proof. Fix a sandpile ordering. If c is a sandpile, i.e., if $c \geq 0$, then the fact that c has a stabilisation follows immediately from the previous lemma and the second property of sandpile orderings.

Given an arbitrary configuration c we can define the sandpile c^+ by

$$c^+(v) = \max\{0, c(v)\}$$

for each $v \in \tilde{V}$. We have just seen that c^+ is stabilisable. Further, by the least action principle, every legal firing sequence for c^+ is finite. Since every legal firing sequence for c is also a legal firing sequence for c^+, it follows that c has no infinite legal firing sequence, and is therefore stabilisable. □

Corollary 2.2.7 (Uniqueness of stabilisation)**.** *Let c be a configuration and $\sigma, \sigma' \geq 0$ be firing scripts corresponding to legal firing sequences for c. Suppose that $\sigma : c \rightsquigarrow \tilde{c}$ and $\sigma' : c \rightsquigarrow \tilde{c}'$, with $\tilde{c}$ and $\tilde{c}'$ both stable. Then $\sigma = \sigma'$ and $\tilde{c} = \tilde{c}'$.*

2.2.4 The identity

The zero configuration is rarely recurrent. Hence, even though $(0 + c)^\circ = c$ for all configurations c, it is seldom the case that 0 is the identity of the sandpile group. It is rather unusual of a group to have a complex identity element which is what we can see is the case here when taking a glance at the images of identity elements on grid graphs such as in Figure 2.1.

Nevertheless we do have a way of calculating the identity but first we need to define what a maximal stable configuration is.

Definition 2.2.8. The *maximal stable configuration* on G is

$$c_{\max} := \sum_{v \in \tilde{V}} (\mathrm{outdeg}(v) - 1)v.$$

Now we can proceed.

Proposition 2.2.9. *The identity of $\mathcal{S}(G)$ is*

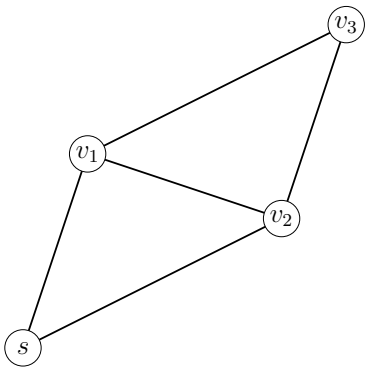
$$(2c_{\max} - (2c_{\max})^\circ)^\circ.$$

Proof. The displayed element is recurrent since

$$(2c_{\max} - (2c^\circ_{\max})^\circ = (c_{\max} + (c_{\max} - (2c_{\max})^\circ))^\circ,$$

and as an element of $\mathbb{Z}\tilde{V}/\tilde{\mathcal{L}}$ it is equal to (the equivalence class of) 0, the identity of $\mathbb{Z}\tilde{V}/\tilde{\mathcal{L}}$. This completes the proof since the isomorphism $\mathcal{S}(G) \to \mathbb{Z}\tilde{V}/\tilde{\mathcal{L}}$ preserves the identity. □

Example 2.2.10. Let G be the diamond graph pictured below, with vertex order v_1, v_2, v_3, s.



Then $c_{\max} = (2, 2, 1)$. One may check - perhaps using the addition table found in Example 2.2.2 - that

$$2c_{\max} = (4, 4, 2) \rightsquigarrow (2, 2, 0) = (2c_{\max})^\circ,$$

with firing script (3,3,4) - meaning we fire three times from v_1 and v_2 and four times from v_3 - and

$$2c_{\max} - (2c_{\max})^\circ = (2, 2, 2) \rightsquigarrow (2, 2, 0) = (2, 2, 0)^\circ.$$

Hence the sandpile identity for G is (2,2,0). This happily matches with what was found before by way of the addition table.

2.2.5 Images of sandpiles on grid graphs

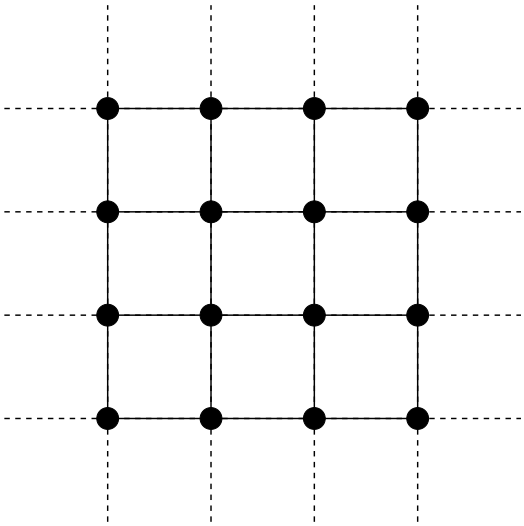
The $m \times n$ sandpile grid graph has non-sink vertices

$$\tilde{V} := \big\{(i, j) \in \mathbb{Z}^2 : 1 \leq i \leq m, 1 \leq j \leq n\big\}$$

with edges between horizontal and vertical neighbours. The vertices (i, j) and (i', j') in $\tilde{V}$ are adjacent if

$$|i - i'| + |j - j'| = 1.$$

Vertices on the boundary of the grid are connected to the sink vertex s. So if $i \in \{1, m\}$ or $j \in \{1, n\}$ then (i, j) is adjacent to the sink. Further, each of the four corner vertices, $(1, 1), (1, n), (m, 1)$, and (m, n), is connected by an extra edge to the sink (making two in total). Thus every non-sink vertex has degree 4. Here is a drawing of the 4×4 sandpile grid:



The dashed boundary edges connect to the sink vertex, which is not pictured here.

Sandpiles on grids display conveniently on computer screens. Each pixel is taken to represent a vertex and the colour of a pixel represents the number of grains of sand on the corresponding vertex - people are just beginning to understand the patterns in these sandpiles.

For instance, imagine placing n grains of sand on a single vertex of an enormous grid graph and stabilising. As n grows, rescale (by a factor of $\sqrt{n}$) so that the image of the resulting sandpile does not change size.

2.3 Tilings

2.3.1 Cayley graphs

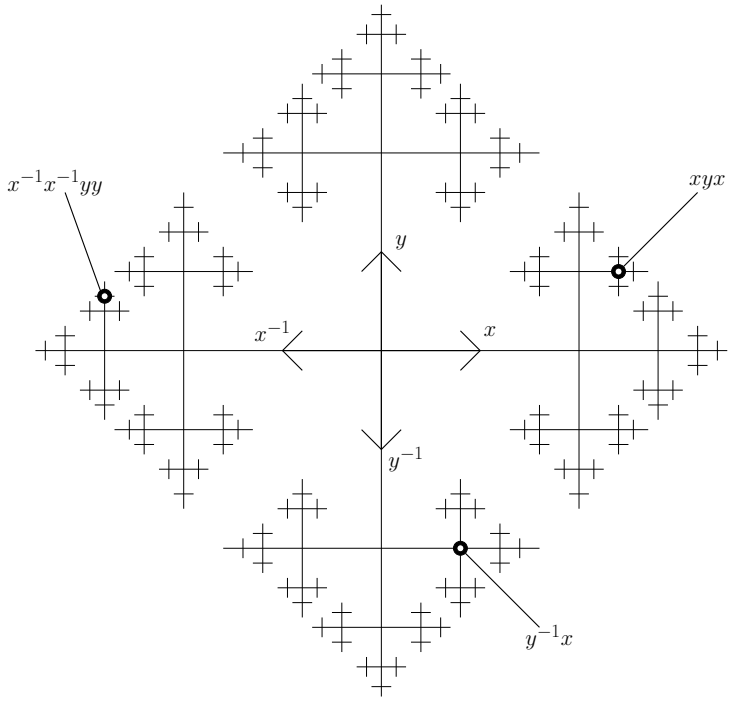


Figure 2.2: The Cayley graph of the free group on two generators x and y with three examples displayed.

In wanting to get the sandpile model working on hyperbolic tilings one thing that needs addressing is the labelling of the tiles. This is particularly important if we hope to be able to write any program to implement firing moves. On a grid graph this is simple, we use the coordinates of the vertices. It is less obvious what should be done in the hyperbolic plane. One way of beginning to solve this problem could be to first focus on ideal tilings, in particular the tilings achieved by using ideal rectangles. From here we can identify this tiling with the Cayley graph of the free group on two generators. Named after Arthur Cayley (1821 - 1895), these graphs are a crucial device in combinatorial and geometric group theory. Shown above is the Cayley graph of the free group on two generators that we need - the generators being x and y - along with some examples of how each point is named.

Definition 2.3.1. Let G be a group and S be a generating set of G. The *Cayley graph* $\Gamma = \Gamma(G, S)$ is an edge-coloured directed graph constructed as follows:

- Each element g of G is assigned a vertex: the vertex set of Γ is identified with G.
- Each element s of S is assigned a colour c_s.
- For every $g \in G$ and $s \in S$, there is a directed edge of colour c_s from the vertex corresponding to g to the one corresponding to gs.

2.3.2 Ideal hyperbolic tilings

The ideal tilings of hyperbolic space appear to be the easiest to work with when it comes to sandpiles since the relations between the individual tiles are simplest.

The Cayley graph of the free group on two generators can be used to represent a tiling of the hyperbolic plane with ideal rectangles. This is because it is the dual graph of the tiling as it has a vertex for each face and an edge for each pair of faces that are separated from each other by an edge - something which can be seen quite clearly in Figure 2.3.

The goal for this section will be to formulate a way of describing the stable state if we place n chips on the central vertex of the four-valent infinite tree. We can see that there will be some restrictions on possible stable finishing points.

Firstly the central vertex will wind up having n (mod 4) chips when all other vertices have stabilised. This is because we can fire the required number of times to stabilise the centre - leaving it with n (mod 4) chips - and then due to the symmetry of the graph whenever a vertex adjacent to the centre is unstable and needs to fire so will the other centre adjacent vertices meaning the centre will gain four chips. Then it can fire again and be left again with n (mod 4) chips. This will continue until full stabilisation is reached across all vertices.

We also notice that because of the symmetry of the four branches out of the central vertex and then the three branches out of each of the vertices next to the centre and then the branches out of the vertices at a distance of two from

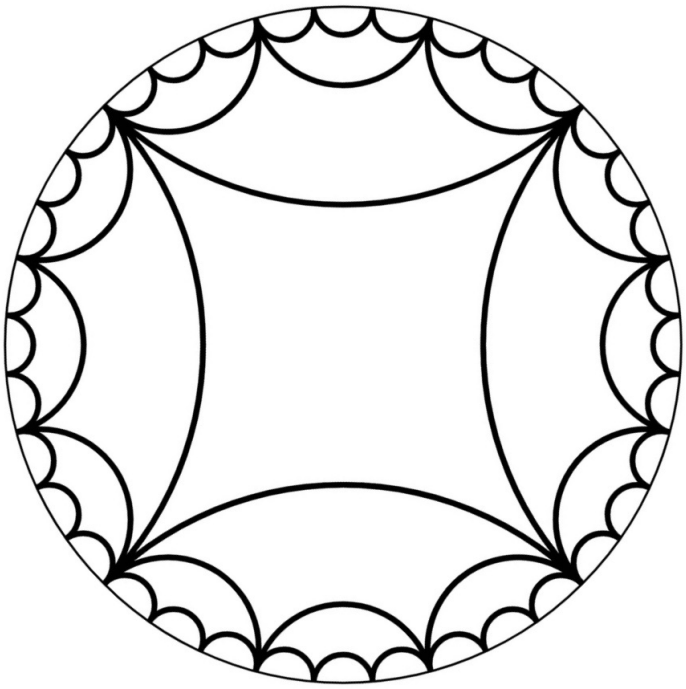


Figure 2.3: The Poincaré disc model of the hyperbolic plane tiled with ideal rectangles - made using an altered version of the code I had written.

the centre and so on and so on, all vertices which are at the same distance from the centre will end up having the same number of chips. For example if a vertex at a distance of seven from the centre finishes with two chips, so will all the other vertices at a distance of seven from the centre.

Next is something interesting that happens once a non-origin vertex gains at least one chip:

Proposition 2.3.2. *If a non-origin vertex has at least one chip at any point during the stabilisation then it will finish non-empty.*

Proof. There are two steps for showing that this is true: firstly it needs to be shown for the vertices connected to the centre and then using induction and the fractal nature of the graph it is enough to show that it is also true for the vertices which are two steps away from the centre.

Starting with n chips on the centre and firing as many times as possible

will leave the neighbouring vertices with $a_1 = \lfloor \frac{n}{4} \rfloor$ chips each. If $a_1 = 0$ then $n < 4$ and they never had any chips so there is nothing to prove. If $1 \leq a_1 \leq 3$ then we are in a stable position and the non-origin vertices that have at some point gained a chip have finished non-empty. If $a_1 = 4$ then these vertices will now fire and will have no chips left which would be a worry if it was not for the fact that now the central vertex has four chips - one from each adjoining - and so can fire and give a chip back to them all, leaving them with one chip each. So whenever the vertices one step from a centre fire they will lose four chips but gain one back meaning they will never be empty again. This extends to all cases where $a_1 > 4$.

When thinking about the vertices that are two steps from the centre - and by induction any number of steps from the centre - we will see something similar. Let a_2 be the number of chips on each of the vertices two steps from the centre. After we fire the central and its connected vertices as much as we need to for those to stabilise either $0 \leq a_2 \leq 3$ in which case we are done or $a_2 \geq 4$ and can itself be fired. Note that this means those vertices between the centre and the ones represented by a_2 will have had chips at some point and so $1 \leq a_1 \leq 3$. Firing now means that a_2 will decrease by four and a_1 will increase by three so $a_1 \geq 4$ and these vertices can be fired, giving a chip back and increasing a_2 by one. It also gives the central vertex four chips which it can give straight back meaning that $a_1 \neq 0$ and $a_2 \neq 0$, as required. If we look at any vertices further away from the centre then we will see an identical structure and so will get the same result. □

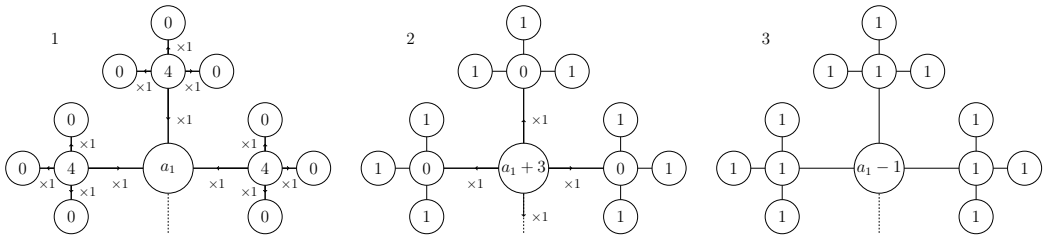


Figure 2.4: A diagram showing how vertices two steps from the centre will always gain back one chip after performing a firing move. Here $1 \leq a_1 \leq 3$ and the centre gains four chips after step 2 and so would fire a chip back to restore a_1 to what it was at the beginning.

Since the total number of chips is conserved we can express the stable position

obtained by placing n chips at the centre in the following way

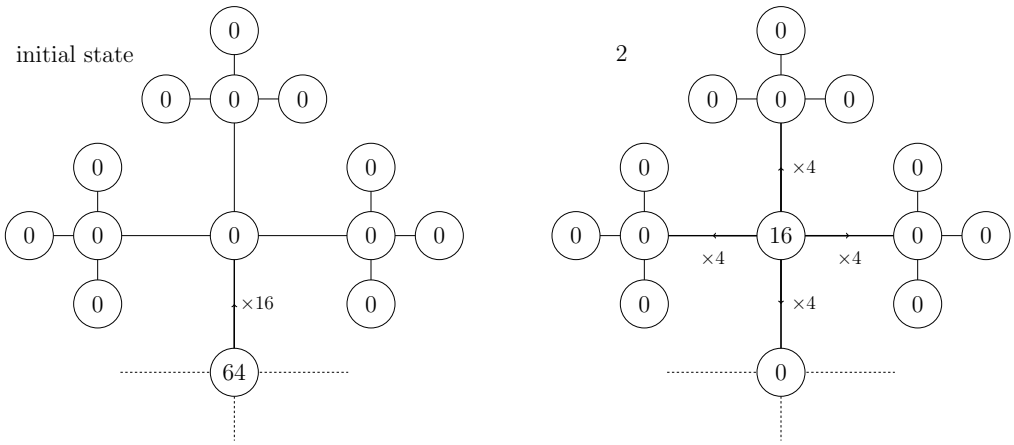
$$n = r + 4(a_1 + 3a_2 + 3^2a_3 + 3^3a_4 + \cdots) \tag{2.1}$$

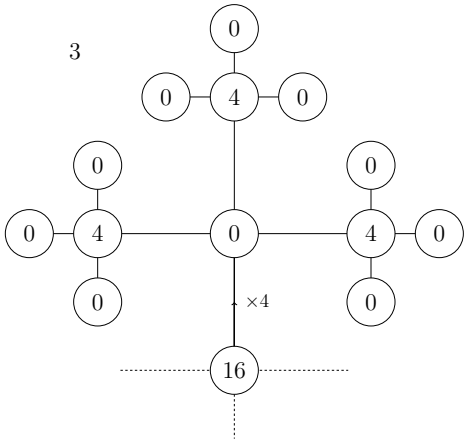
where n is the initial number of chips, r is the number of chips on the central vertex, and a_i is the number of chips on each (and every) vertex at a distance of i from the centre. The variables n, r, and the a_is all take integer values. The multiplication by four is required because of the four symmetrical branches. The powers of three come from the branches out of the vertices that get further and further from the centre - there are three extending out of the vertices adjacent to the centre and then a further three from each of these and so on and so on. Clearly $r = n \pmod 4$ so $0 \leq r \leq 3$ and as we have said already, if at any point we fire from a_{i-1} meaning $a_i > 0$ we will finish with $1 \leq a_i \leq 3$.

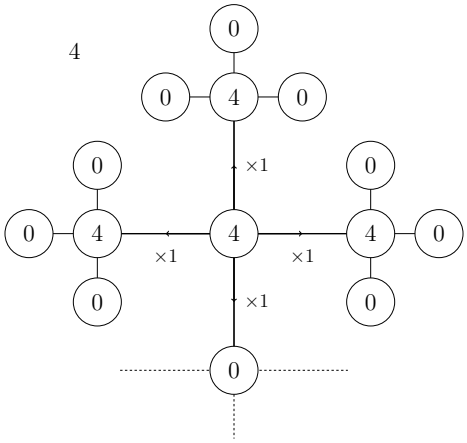
Along with the fact that each sandpile system has a unique stable state this looks promising. We hope that there is a unique representation for each n in the form of (2.1) and that this can be found in a manner simpler than having to perform every firing move needed. In order to show that this unique representation exists it would be enough to show that there is a unique way to write $q = a_1 + 3a_2 + 3^2a_3 + 3^3a_4 + \cdots$ for all $q \geq 1$.

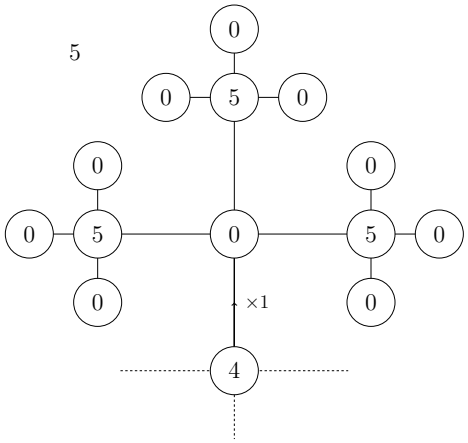
Before going any further here is an explicit example:

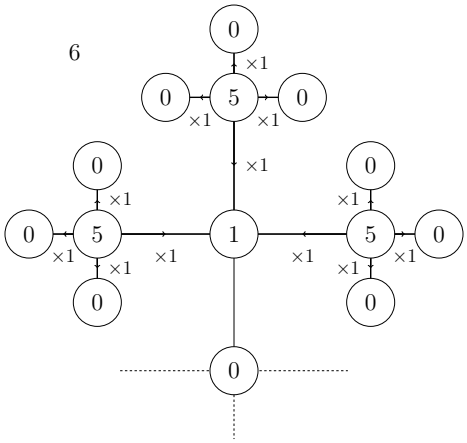
Example 2.3.3. Using the Cayley graph of the free group on two generators as our tree we can place 64 chips on the central vertex and see how it stabilises.

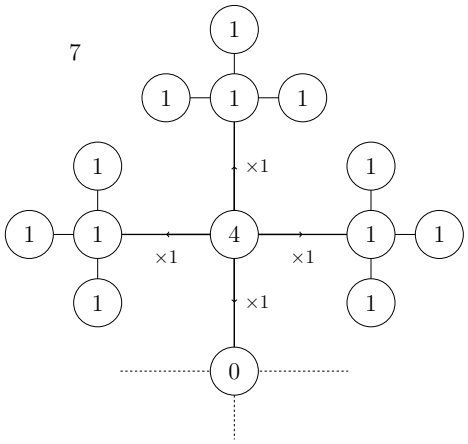


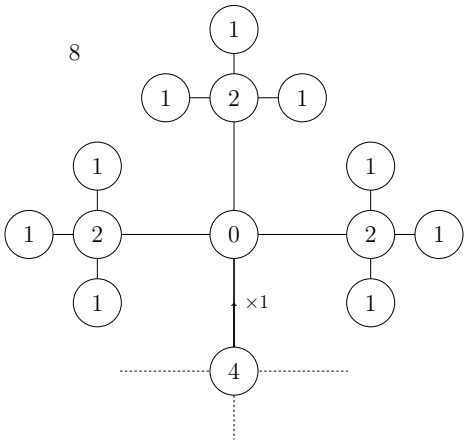

3
0
0 4 0
0 0
0 4 0 4 0
0 0
×4
16

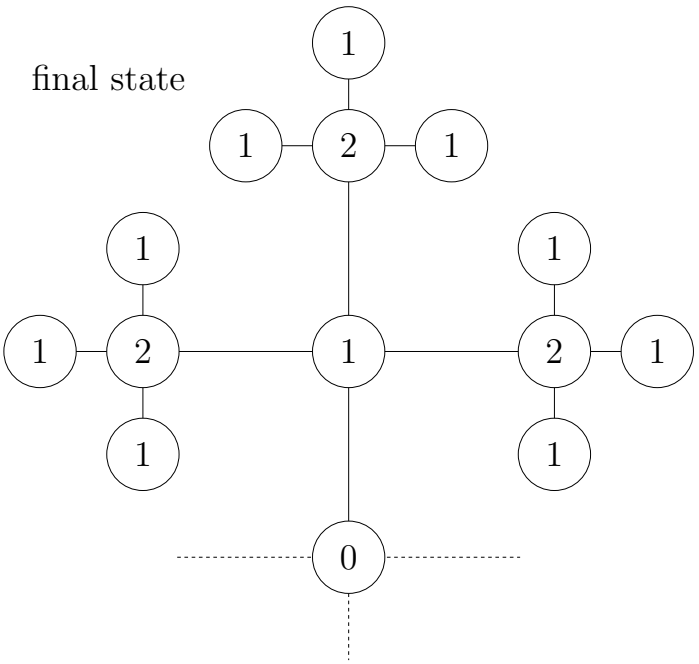

4
0
0 4 0
0 ×1 0
0 4 ×1 4 ×1 4 0
0 ×1 0
0


5
0
0 5 0
0 0
0 5 0 5 0
0 0
×1
4


6
0
×1
0 5 0
×1 ×1
0 ×1 0
×1 ×1
0 5 ×1 1 ×1 5 0
×1 ×1 ×1 ×1 ×1 ×1
0
0


7
1
1 1 1
1 ×1 1
1 1 4 1 1
×1 ×1
1 ×1 1
0


8
1
1 2 1
1 1
1 2 0 2 1
1 1
×1
4



A quick way to check that this is valid is to use expression (2.1). We have $r = 0$, $a_1 = 1$, $a_2 = 2$, $a_3 = 1$ and $a_i = 0$ for all other i. Hence,

$$r + 4(a_1 + 3a_2 + 3^2 a_3) = 0 + 4(1 + 3 \cdot 2 + 3^2 \cdot 1) = 4(1 + 6 + 9) = 4 \cdot 16 = 64,$$

as we should have been expecting.

It is clear to see that solving these, while fiddly but not too difficult to do by hand, can be rather cumbersome - even more so as we increase the initial number of chips placed onto the graph.

Using a colour scheme where blue corresponds to tiles with zero chips, lilac to one, green to two, and beige to three (which is not really needed in this case) we obtain the tiling shown in Figure 2.5.

This is also a rather nice example in particular since it feeds well into the idea of a relation to a ternary numeral system. We begin with 64 chips and so each branch has 16 chips of its own. Since $16 = 1 \cdot 3^2 + 2 \cdot 3^1 + 1 \cdot 3^0$, it is written as 121 in base three - the usual notation for this is $16 = 121_3$. This matches the stable state we ended up with when we ignore the central vertex and let the digits 121_3 be paired with the string $a_3 a_2 a_1$. It would be very nice if all we had to do to find the numbers needed for each tiling was to find

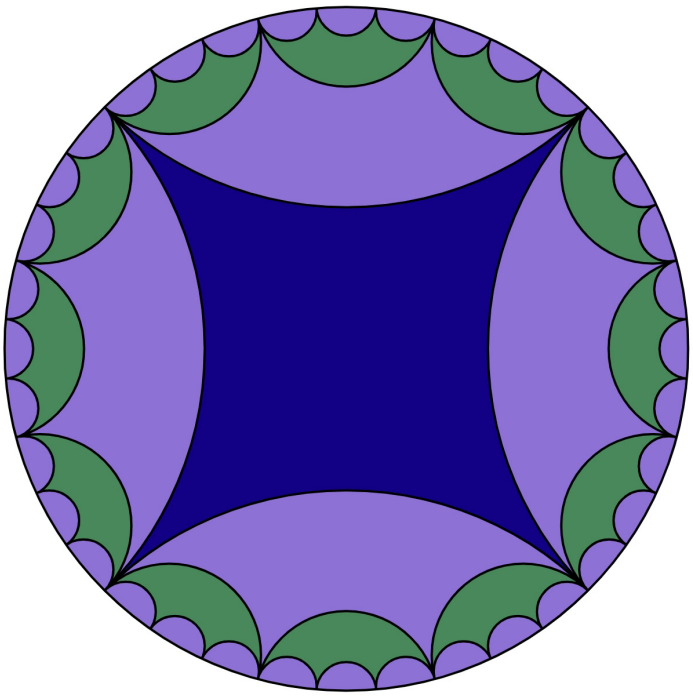


Figure 2.5: The ideal rectangle tiling obtained when we place 64 chips on the central vertex and use blue to colour tiles with zero chips, lilac to one, green to two, and beige to three.

the base three representation of $\lfloor \frac{n}{4} \rfloor$ and while this does work in many cases things are unfortunately not always so simple.

In the cases where the base 3 representation of the decimal number $\lfloor \frac{n}{4} \rfloor$ contains no zeroes this is in fact enough. When the ternary number does contain at least one zero there is a little more work to do. Take $102_3 = 11$ for example. We have $1 \cdot 3^2 + 0 \cdot 3^1 + 2 \cdot 3^0$ which is the same as borrowing from the next step up from where the zero is and instead writing $0 \cdot 3^2 + 3 \cdot 3^1 + 2 \cdot 3^0$. Below is a table featuring some more examples of how this borrowing system can convert standard base 3 representations to the slightly deformed versions we desire for our tilings:

decimal	3^5	3^4	3^3	3^2	3^1	3^0	ternary	tiling $(a_1a_2a_3\cdots)$
11				1	0	2	102_3	-
11				0	3	2	-	23
15				1	2	0	120_3	-
15				1	1	3	-	311
173		2	0	1	0	2	20102_3	-
173		1	3	1	0	2	-	-
173		1	3	0	3	2	-	-
173		1	2	3	3	2	-	23321
173		2	0	1	0	2	20102_3	-
173		2	0	0	3	2	-	-
173		1	3	0	3	2	-	-
173		1	2	3	3	2	-	23321
378	1	1	2	0	0	0	11200_3	-
378	1	1	1	3	0	0	-	-
378	1	1	1	2	3	0	-	-
378	1	1	1	2	2	3	-	322111

The two different paths for finding the tiling number for 173 shows that applying the method does not have to follow a specific pattern though my personal preference is to work down from the largest indices that require attention.

Now we can combine all of the above to describe the tilings in a more formal setting:

Theorem 2.3.4. *For any number of chips $n \in \mathbb{N}$ placed on the central vertex of the Cayley graph of the free group on two generators there is a unique representation*

$$n = r + 4(a_1 + 3a_2 + 3^2a_3 + 3^3a_4 + \cdots)$$

of the stable state where r is taken to be the number of chips on the central vertex and a_i to be the number of chips on all vertices at a distance of i from the centre with $0 \leq a_i \leq 3$ for all i and if $a_j > 0$ then so is a_k for all $k < j$.

Proof. Let $r = n \pmod 4$. With some substitution and rearranging we can see that

$$\frac{n - n \pmod 4}{4} = a_1 + 3a_2 + 3^2a_3 + 3^3a_4 + \cdots .$$

As $n - n \pmod 4$ will always be divisible by four the left hand side of the above equation will always be an integer. Then for the theorem to be proven we need that any integer can be uniquely represented by the right hand side under the stated constraints.

Let $q = a_1 + 3a_2 + 3^2a_3 + 3^3a_4 + \cdots$. This looks similar to how numbers are represented in a ternary numeral system where the base is three and we know that every $q \in \mathbb{Z}$ has a unique base three representation. However this will not always remain valid under our constraints as the base three number could have $a_k = 0$ and $a_j > 0$ with $k < j$. It can be adapted though.

Take the standard base three representation of q. If it contains no zeroes then we are done and the values of $a_1, a_2, a_3, \cdots$ are equal to the base three number when written as $\cdots a_3a_2a_1$. For example $476_{10} = 122122_3$ so the six digits can be identified with the string $a_6a_5a_4a_3a_2a_1$ and we take $a_i = 0$ for all $i > 6$ - once we have one a_i equal to zero all of the a_is with higher subscripts will also be zeroes. If the standard base three representation does contain zeroes however we can follow the borrowing algorithm described above until all zeroes are removed and we are left with a string of digits containing only ones, twos, and threes. Each digit can be paired up with the appropriate a_i in the same way as before - taking the reverse order.

Since the number of chips in the graph is conserved we can declare that this representation is unique because changing any of the values of r or the a_is would increase or decrease the total number of chips and so it would no longer be equal to n. □

Therefore given any number n of chips placed on the central vertex of the Cayley graph representing the symmetric tiling of the Poincaré disc using ideal rectangles we can (somewhat) quickly determine how the graph will look when stabilised by expressing n in the unique way described above.

Here is one final example just to make it blindingly obvious how much faster and easier this method is than implementing each and every firing move we would need to in order to stabilise the system.

Example 2.3.5. Let $n =$ four billion. Dropping all of these chips on the central vertex will mean it can fire and give each of the four branches a billion chips to play with. Four billion is divisible by four so $r = 0$. Now to determine

values for the a_is we begin by getting the base 3 representation of one billion:

$$1,000,000,000 = 2120200200021010001_3.$$

So the borrowing goes as follows:

2	1	**2**	**0**	2	0	0	2	0	0	0	2	1	0	1	0	0	0	1
2	1	1	3	**2**	**0**	0	2	0	0	0	2	1	0	1	0	0	0	1
2	1	1	3	1	**3**	**0**	2	0	0	0	2	1	0	1	0	0	0	1
2	1	1	3	1	2	3	**2**	**0**	0	0	2	1	0	1	0	0	0	1
2	1	1	3	1	2	3	1	**3**	**0**	0	2	1	0	1	0	0	0	1
2	1	1	3	1	2	3	1	2	**3**	**0**	2	1	0	1	0	0	0	1
2	1	1	3	1	2	3	1	2	2	3	2	**1**	**0**	1	0	0	0	1
2	1	1	3	1	2	3	1	2	2	3	**2**	**0**	3	1	0	0	0	1
2	1	1	3	1	2	3	1	2	2	3	1	3	3	**1**	**0**	0	0	1
2	1	1	3	1	2	3	1	2	2	3	1	3	**3**	**0**	3	0	0	1
2	1	1	3	1	2	3	1	2	2	3	1	3	2	3	**3**	**0**	0	1
2	1	1	3	1	2	3	1	2	2	3	1	3	2	3	2	**3**	**0**	1
2	1	1	3	1	2	3	1	2	2	3	1	3	2	3	2	2	3	1

Hence using (2.1) to express the number of chips on each tile we have:

r	a_1	a_2	a_3	a_4	a_5	a_6	a_7	a_8	a_9	a_{10}	a_{11}	a_{12}	a_{13}	a_{14}	a_{15}	a_{16}	a_{17}	a_{18}	a_{19}
0	1	3	2	2	3	2	3	1	3	2	2	1	3	2	1	3	1	1	2

and $a_i = 0$ for all $i > 19$.

2.4 Next steps

The only tiling worked on here is the one using ideal rectangles. It seems as though we could reach similar results for other ideal tilings using infinite trees with different valencies and deformations of other numeral systems - using binary numbers for the tiling using ideal triangles, for instance.

The symmetric spaces of infinite trees where every vertex has the same degree are examples of CAT(0) spaces - also known as Hadamard spaces, named after Jacques Hadamard (1865 - 1963). Another example of a Hadamard space is the Poincaré disc. So in a sense these symmetric ideal tilings of the Poincaré disc are the most hyperbolic ones possible but this will not be looked into in any further detail.

In any case dropping chips on the central vertex of a symmetric (either reflective or rotational or both) ideal tiling will result in a stable state which is also symmetric. This gives us nice patterns that can be deciphered simply but it is unclear what would happen when we try applying these sandpile models to different tilings which are not made using ideal shapes or do not contain symmetries.

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Appendix A

Using JavaScript and p5.js

As part of the dissertation project I thought it would be interesting to write some code that could draw the hyperbolic tilings produced by dropping large numbers of chips on graphs and letting them stabilise. Unfortunately I did not quite manage to make it this far but I knew that in order to draw the tilings I would need to be able to draw geodesics and so this is where I began and I did create a few things along the way.

I also wanted to be able to share what I was making and allow for user interaction. The language I have used most and become most adept at - though by no means an expert - in past years at the University of Sheffield is Python. It quickly became apparent however that the best language to use for this would be JavaScript, something I had never worked with before. To overcome this obstacle I followed the introductory courses offered by Codecademy on JavaScript [5] and p5.js [6] which is used to create drawings.

After looking at several options it seemed that the best way to share what I had made to be would be to publish a GitHub page and here is the link:

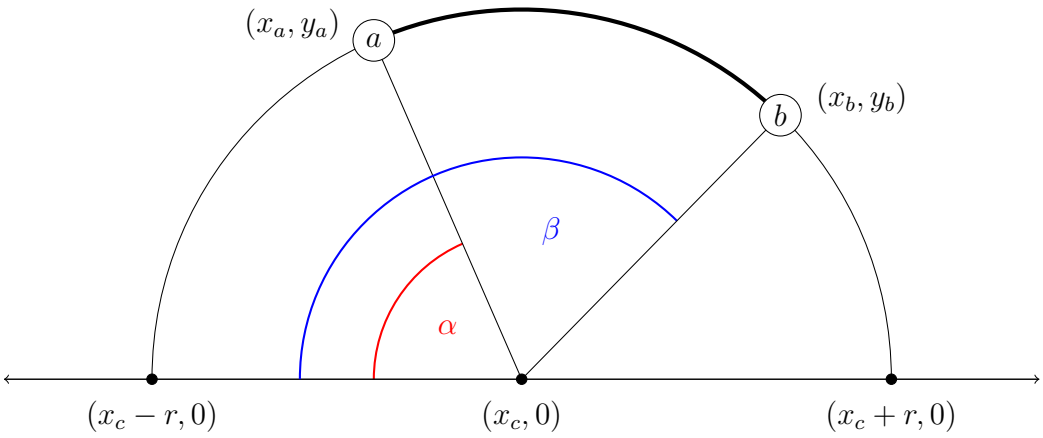
cllmhntngtn.github.io/hyperbolicgeodesics

There are still a few bugs and some incomplete tidying up but for the most part what I have runs quite smoothly. What follows is some discussion about the activity occuring under the hood.

A.1 Drawing geodesics in $\mathbb{H}$

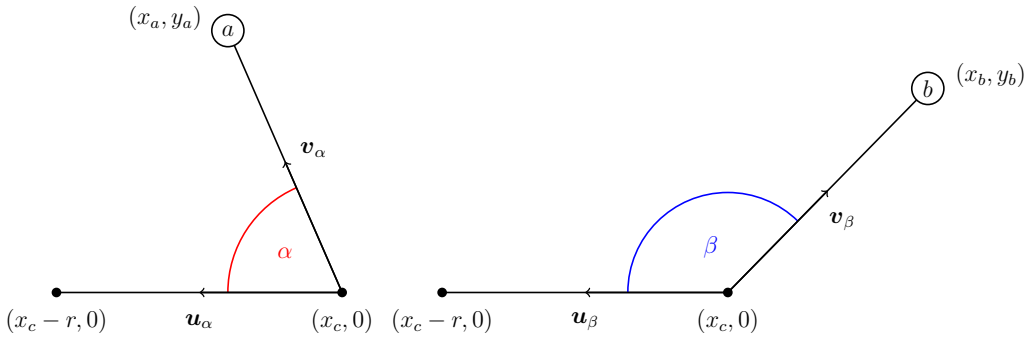
We saw in the first chapter that the geodesics in $\mathbb{H}$ are the semi-circles orthogonal to the real axis and the vertical straight lines. Further to this the candidates for geodesics in the upper half-plane can be described uniquely by their end points in $\partial\mathbb{H}$. Semi-circles orthogonal to $\mathbb{R}$ have two end points in $\mathbb{R}$ and vertical lines have one end point in $\mathbb{R}$ and the other at ∞. After learning some JavaScript and using some Euclidean geometry it was possible to code an interactive program wherein a user can input values for any two points in $\mathbb{H}$ and receive a drawing of the unique geodesic passing between them.

After the two points have been inputted the program will first determine whether they have the same real part in which case the geodesic is the vertical line between them. If not then the centre of the semi-circle is calculated by determining the point on the real axis which is equidistant to the two given points. The radius of the relevant semi-circle is then the distance between the centre and either of the given points and so the two end points in $\mathbb{R}$ can be found by moving the distance of the radius along the real axis in both the positive and negative directions. From here the semi-circle can be drawn. An extra touch I applied was to colour the arc between the two points differently to the rest of the semi-circle to highlight the path between the two points along the larger geodesic.



In order to calculate the two angles needed we can consider the radii relevant

to each angle as vectors and use some vector calculus to compute the angle between them.



The dot product of two Euclidean vectors $\boldsymbol{u}$ and $\boldsymbol{v}$ is defined by

$$\boldsymbol{u} \cdot \boldsymbol{v} = ||\boldsymbol{u}|| \; ||\boldsymbol{v}|| \cos\theta$$

where θ is the angle between them. Rearranging and writing out each calculation explicitly gives us the following formulae for calculating the angles we need

$$\alpha = \arccos\left[\frac{-r(x_a - x_c)}{r\sqrt{(x_a - x_c)^2 + y_a^2}}\right], \qquad \beta = \arccos\left[\frac{-r(x_b - x_c)}{r\sqrt{(x_b - x_c)^2 + y_b^2}}\right].$$

When it came to writing some code using JavaScript and p5.js one of the first things I needed to do was to establish where the origins of the axes were, calling these `xOriginUHP` and `yOriginUHP`. Then I decided on how large I wanted each step in the x and y directions to be, calling these `xStepUHP` and `yStepUHP`, respectively.

The values of the coordinates of two points inputted by the user are assigned to variables named `x[0]`, `y[0]` and `x[1]`, `y[1]`. As more points are inputted and more geodesics drawn the indices increase to 2 and 3, 4 and 5, and so on and so on and they are added to a new array.

Here is the code to calculate what is needed to draw the geodesics:

```
if (x[0] && y[0] && x[1] && y[1]) {
  // calculating the relevant semi-circle
  var circleCentre = (x[0] ** 2 + y[0] ** 2 - x[1] ** 2 - y
    [1] ** 2) / (2 * x[0] - 2 * x[1]);
```

```
  var circleRadius = dist(x[0], y[0], circleCentre, 0);
  // calculating angles for the arc along the semi-circle
  var alpha = acos((- circleRadius * (x[0] - circleCentre)) /
    (circleRadius * sqrt((x[0] - circleCentre) ** 2 + y[0] **
    2)));
  var beta = acos((- circleRadius * (x[1] - circleCentre)) /
   (circleRadius * sqrt((x[1] - circleCentre) ** 2 + y[1] **
   2)));
}
```

Here is the code which draws the geodesics:

```
// drawing the geodesics
if (x[0] && y[0] && x[1] && y[1]) {
      strokeWeight(6);
      strokeCap(SQUARE);
  noFill();
      if (x[0] === x[1]) {
          stroke('darkgray');
          line(xOriginUHP + x[0] * xStepUHP, yOriginUHP,
   xOriginUHP + x[1] * xStepUHP, 0);
          stroke('dodgerblue');
          line(xOriginUHP + x[0] * xStepUHP, yOriginUHP - y
   [0] * yStepUHP, xOriginUHP + x[1] * xStepUHP, yOriginUHP -
    y[1] * yStepUHP);
      } else if (x[0] < x[1]) {
          stroke('darkgray');
          arc(xOriginUHP + circleCentre * xStepUHP,
   yOriginUHP, 2 * circleRadius * xStepUHP, 2 * circleRadius
   * yStepUHP, PI, 0, OPEN);
          stroke('dodgerblue');
          arc(xOriginUHP + circleCentre * xStepUHP,
   yOriginUHP, 2 * circleRadius * xStepUHP, 2 * circleRadius
   * yStepUHP, PI + alpha, PI + beta, OPEN);
      } else if (x[0] > x[1]) {
          stroke('darkgray');
          arc(xOriginUHP + circleCentre * xStepUHP,
   yOriginUHP, 2 * circleRadius * xStepUHP, 2 * circleRadius
   * yStepUHP, PI, 0, OPEN);
          stroke('dodgerblue');
          arc(xOriginUHP + circleCentre * xStepUHP,
   yOriginUHP, 2 * circleRadius * xStepUHP, 2 * circleRadius
   * yStepUHP, PI + beta, PI + alpha, OPEN);
      }
}
```

Figure A.1: Examples of geodesics drawn in the upper half-plane.

A.2 Drawing geodesics in $\mathbb{D}$

Also in the first chapter the geodesics in $\mathbb{D}$ were described as the arcs of circles and diameters in $\mathbb{D}$ that meet $\partial\mathbb{D}$ orthogonally. As suggested there it is easier to transfer the results from the upper half-plane directly to the Poincaré disc. Making use of the `complex.js` library for using complex numbers in JavaScript along with the map $h : \mathbb{H} \to \mathbb{D}$ defined by

$$h(z) = \frac{z - i}{iz - 1}$$

and some more Euclidean geometry I was able to write code analogous to that which works in the upper half-plane to draw geodesics in the Poincaré disc.

Here is the code which converts points in the upper half-plane to points in the disc:

```
// converting the points in the half-plane to points in the
   disc
var z_1 = new Complex(x[0], y[0]);
var hz_1 = (z_1.sub(Complex['I'])).div(Complex['I'].mul(z_1).
   sub(1));
```

Using the above formula when the geodesic in the upper half-plane is a semi-circle we can convert the two points themselves and the two points where the end points meet the real axis into two points in the Poincaré disc and two points on the boundary, respectively. Since a circle can be defined by any three non-collinear points we can from here choose three of these four points to be (a_x, a_y), (b_x, b_y), and (c_x, c_y) and use the following formulae to find the centre of the circle $u = (u_x, u_y)$ on which the geodesic arc lies

$$\begin{aligned}
d &= 2\left[a_x(b_y - c_y) + b_x(c_y - a_y) + c_x(a_y - b_y)\right], \\
u_x &= \frac{1}{d}\left[(a_x^2 + a_y^2)(b_y - c_y) + (b_x^2 + b_y^2)(c_y - a_y) + (c_x^2 + c_y^2)(a_y - b_y)\right], \\
u_y &= \frac{1}{d}\left[(a_x^2 + a_y^2)(c_x - b_x) + (b_x^2 + b_y^2)(a_x - c_x) + (c_x^2 + c_y^2)(b_x - a_x)\right].
\end{aligned}$$

Here is the code which calculates all that is needed to draw the arcs of circles in $\mathbb{D}$ that meet $\partial\mathbb{D}$ at right-angles:

```
// converting the points where the semi-circle meets the real
    line to points on the disc boundary
var circleCentre = (x[0] ** 2 + y[0] ** 2 - x[1] ** 2 - y[1]
   ** 2) / (2 * x[0] - 2 * x[1]);
var circleRadius = dist(x[0], y[0], circleCentre, 0);
// left point
var leftPointX = circleCentre - circleRadius;
var leftPointUHP = new Complex(leftPointX, 0);
var leftPointPD = (leftPointUHP.sub(Complex['I'])).div(
   Complex['I'].mul(leftPointUHP).sub(1));
// rightPoint
var rightPointX = circleCentre + circleRadius;
var rightPointUHP = new Complex(rightPointX, 0);
var rightPointPD = (rightPointUHP.sub(Complex['I'])).div(
   Complex['I'].mul(rightPointUHP).sub(1));
// calculating the centre of the semi-circle which contains
   the geodesic
var a = [xOriginPD + hz_1.re * discRadius, yOriginPD - hz_1.
   im * discRadius];
var b = [xOriginPD + hz_2.re * discRadius, yOriginPD - hz_2.
   im * discRadius];
var c = [xOriginPD + leftPointPD.re * discRadius, yOriginPD -
    leftPointPD.im * discRadius];
var d = 2 * (a[0] * (b[1] - c[1]) + b[0] * (c[1] - a[1]) + c
   [0] * (a[1] - b[1]));
var semiCentreX = ((a[0]**2 + a[1]**2) * (b[1] - c[1]) + (b
   [0]**2 + b[1]**2) * (c[1] - a[1]) + (c[0]**2 + c[1]**2) *
   (a[1] - b[1])) / d;
var semiCentreY = ((a[0]**2 + a[1]**2) * (c[0] - b[0]) + (b
   [0]**2 + b[1]**2) * (a[0] - c[0]) + (c[0]**2 + c[1]**2) *
   (b[0] - a[0])) / d;
var semiRadius = dist(semiCentreX, semiCentreY, a[0], a[1]);
```

Now the same method as before can be used to determine where to draw the highlighting blue arc. In the case that the geodesic in $\mathbb{H}$ is a vertical line the single point at which it meets the real axis can be taken to be the third point needed to calculate and draw the unique circle wanted for the disc.

If the two points happen to lie on a diameter of the disc and so are collinear and give no circle to be found we can check to see if a straight line between the two points which lie on the boundary has the same length of the diameter of

the disc and if so draw a straight line between them which will pass through the two inputted points.

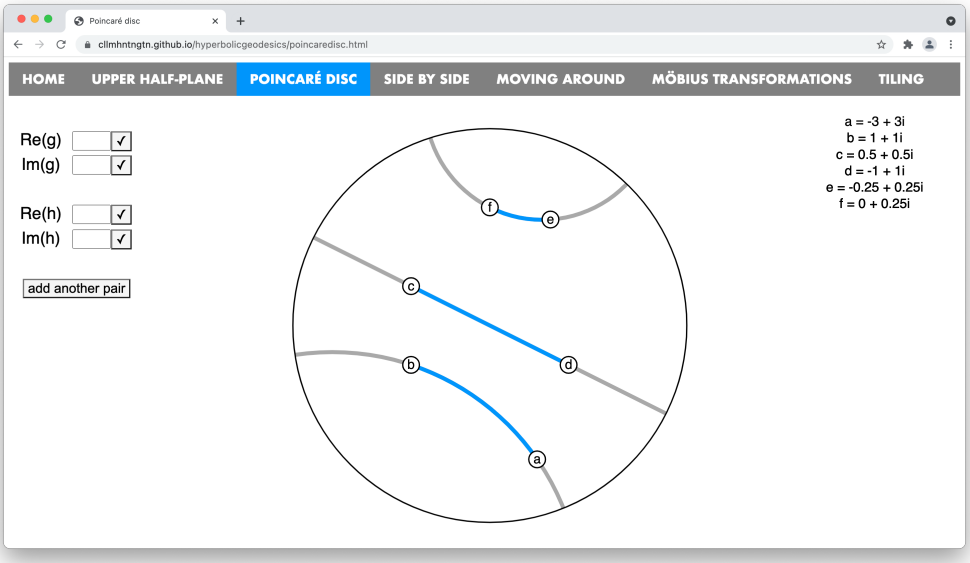


Figure A.2: Examples of geodesics drawn in the Poincaré disc.

A.3 Drawing side by side

As well as being able to draw the geodesics in each of the models separately I thought it would be nice to be able to present them side by side so you can see how the points and geodesics correspond to one another. After reconfiguring some of the aesthetic details of the code this was achieved.

A.4 Interactivity

As well as writing something that would allow a user to specify any two points and see the geodesic between them in both models of hyperbolic space I thought it would be fun to have something where points can be dragged around the screen while the geodesic between them moves with the points.

I already had most of the code I needed and from there had to create the

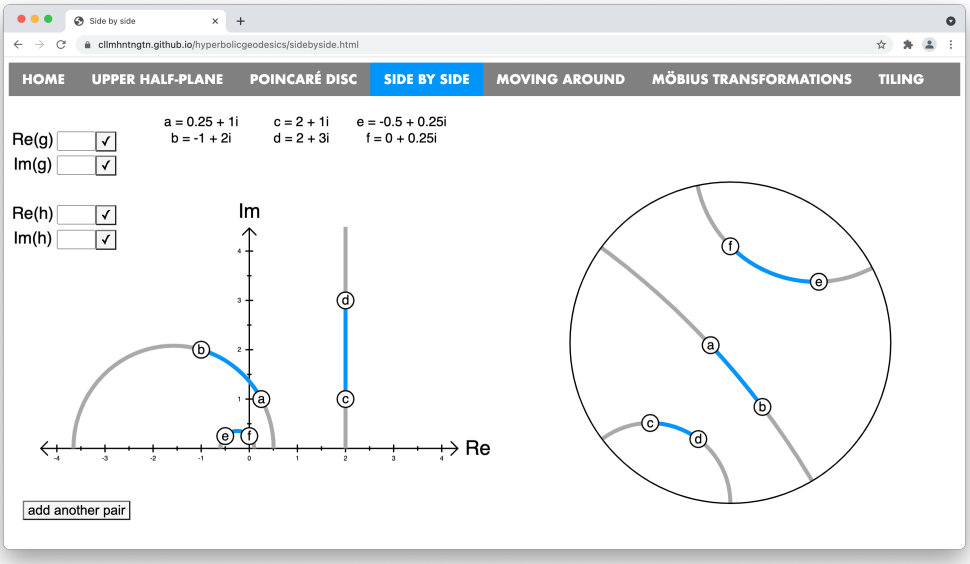


Figure A.3: Examples of geodesics drawn in both the upper half-plane and the Poincaré disc for comparison.

dynamic aspects. The first thing to do was to initialise a few variables related to the positions of the vertices and the position of the mouse when hovering over and interacting with the webpage:

```
// is the mouse over a?
var rolloverA = false;
// is a being dragged?
var draggingA = false;
// is the mouse over b?
var rolloverB = false;
// is b being dragged?
var draggingB = false;

// locations and diameter
var xA, yA, xB, yB, d;
// mouseclick offset
var offsetXA, offsetYA, offsetXB, offsetYB;
```

Then we need to define functions that will work when the mouse is pressed. The code below determines if a vertex is being clicked on and if this is the

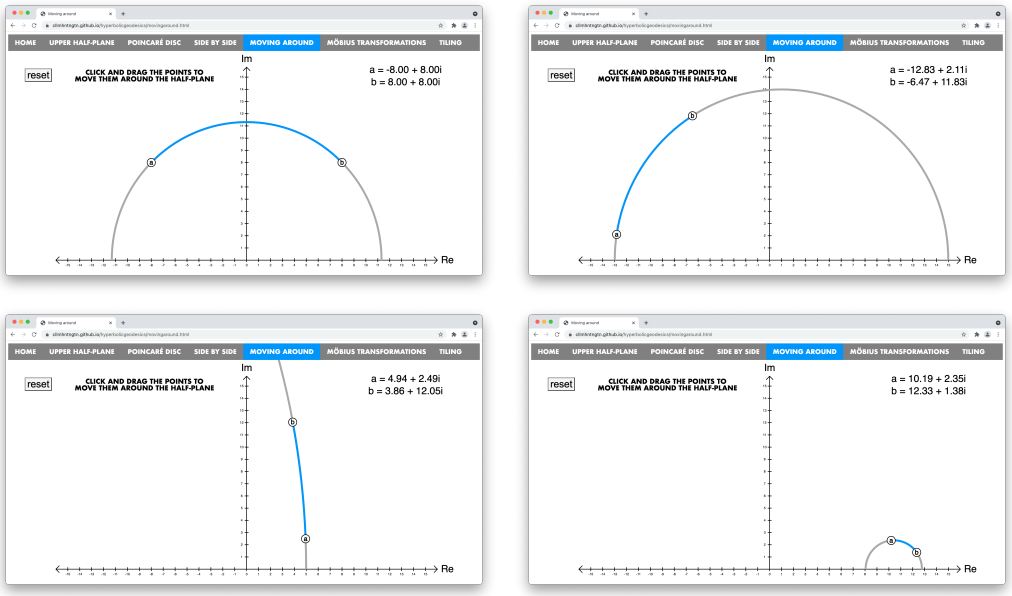


Figure A.4: On this interactive page the user can click and drag the points to move them around the upper half-plane. Admittedly screenshots are not the ideal way of showing this.

case it will set `draggingA` or `draggingB` to be true and keep track of where that vertex is being dragged to. Once the mouse is released it will no longer change the properties of the vertices and will leave them as they were at the point of release:

```
function mousePressed() {
  // did I click on a?
  if (mouseX > xA - d / 2 && mouseX < xA + d / 2 && mouseY >
   yA - d / 2 && mouseY < yA + d / 2) {
    draggingA = true;
    // if so, keep track of relative location of click to
   centre of a
    offsetXA = xA - mouseX;
    offsetYA = yA - mouseY;
  // did I click on b?
  } else if (mouseX > xB - d / 2 && mouseX < xB + d / 2 &&
   mouseY > yB - d / 2 && mouseY < yB + d / 2) {
    draggingB = true;
    // if so, keep track of relative location of click to
   centre of b
```

```
    offsetXB = xB - mouseX;
    offsetYB = yB - mouseY;
  }
}

function mouseReleased() {
  // stop dragging
  draggingA = false;
  draggingB = false;
}
```

This information is simultaneously used to adjust the position of the relevant vertex in the canvas:

```
// adjust location of a if being dragged
if (draggingA) {
  xA = mouseX + offsetXA;
  yA = mouseY + offsetYA;
}

// adjust location of b if being dragged
if (draggingB) {
  xB = mouseX + offsetXB;
  yB = mouseY + offsetYB;
}
```

The code for drawing geodesic between the two points is constantly running on a loop so as soon as one vertex is moved the new geodesic is drawn immediately. This means a user can change the positions of the vertices and instantly see how the size and position of the entire semi-circle - or vertical straight line, though the shape will not change much here - transforms as well as the arc between the two points representing the shortest path.

A.5 Möbius transformations

Applying Möbius transformations to two points was fairly painless using `complex.js` and the standard JavaScript mathematics operations.

```
var z_1 = new Complex(x[0], y[0]);
var z_2 = new Complex(x[1], y[1]);

var mobz_1 = (((z_1.mul(mobiusA)).add(mobiusB))).div(((z_1.
   mul(mobiusC)).add(mobiusD)));
```

```
var mobz_2 = (((z_2.mul(mobiusA)).add(mobiusB))).div(((z_2.
    mul(mobiusC)).add(mobiusD)));

var newReA = mobz_1.re;
var newImA = mobz_1.im;
var newReB = mobz_2.re;
var newImB = mobz_2.im;
```

These new variables are then used to draw the new geodesics.

I did have some trouble with making it allow the visitor decide the original positions of the vertices and could not find a satisfying way of fixing it so for now the user can only choose the values of a, b, c, and d to be used in the Möbius transformation.

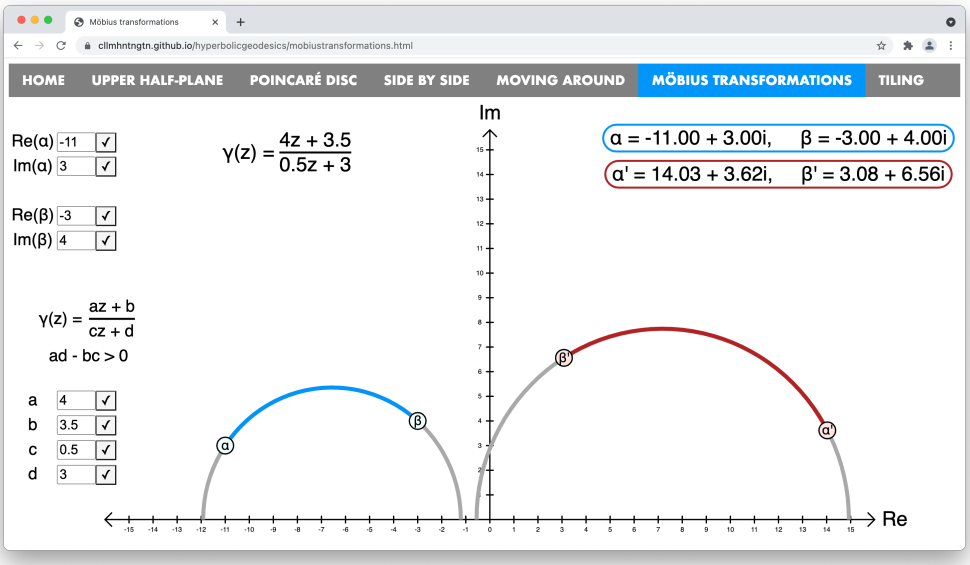


Figure A.5: A first example of a Möbius transformation.

Using Equation 1.1 to calculate the hyperbolic distance between two points we can confirm that the blue arcs and red arcs in the examples in Figures A.5 and A.6 have the same length despite how they may appear to our Euclidean eyes. This was done for Figure A.6 in Example 1.2.5.

Playing around with the Möbius transformations also reaffirms what is stated

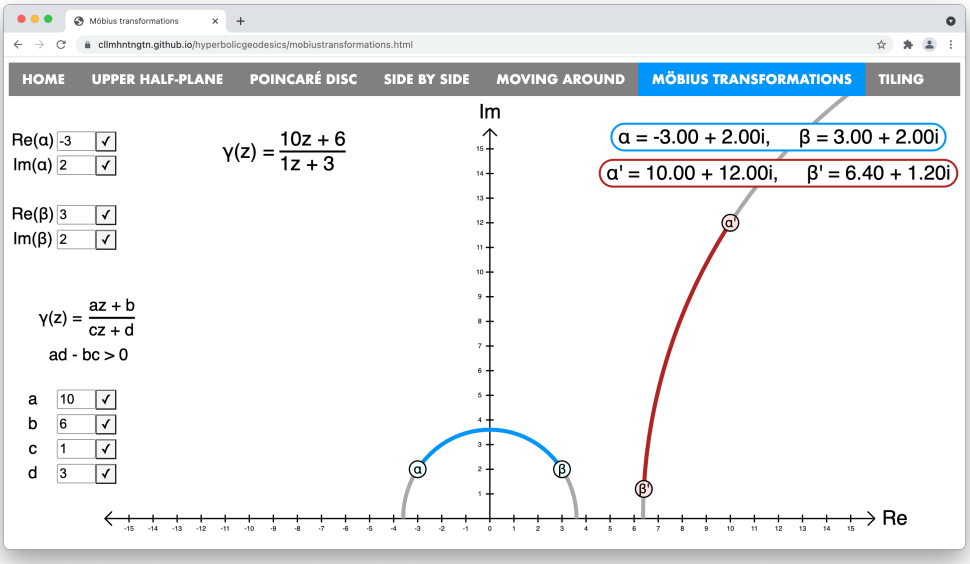


Figure A.6: A second example of a Möbius transformation.

in Proposition 1.3.1: that applying Möbius transformations to either semi-circles orthogonal to the real axis or vertical straight lines will return either semi-circles orthogonal to the real axis or vertical straight lines.

A.6 Tiling

The final thing I managed to do was to create something that would take an input of a number of chips and draw the tiling obtained by dropping said chips on the central vertex of the Cayley graph of the free group on two generators and letting the system stabilise in accordance with the sandpiles introduced in Chapter 2.

I feel as though I had a few of the ingredients needed to get there but was not fully able to find a particularly elegant method of achieving this. Hence I was only able to draw tiles representing vertices up to (and including) a distance of three from the centre and was only able to get it to work for a total of up to (and including) 159 chips - the maximal stable configuration on vertices that are no more than a distance of three from the centre

To begin with I chose some colours to represent the various numbers of chips. Here are the ones I chose for no reason other than I thought they looked nice together:

```
const zero = 'darkblue';
const one = 'mediumpurple';
const two = 'seagreen';
const three = 'bisque';
```

The rather crude method I implemented to determine the pattern was to simply list all of the patterns obtained as the number of chips changes, like this:

```
...
if (n == 130) {chipPattern = [two,two,one,three];}
if (n == 131) {chipPattern = [three,two,one,three];}
if (n == 132) {chipPattern = [zero,three,one,three];}
if (n == 133) {chipPattern = [one,three,one,three];}
if (n == 134) {chipPattern = [two,three,one,three];}
if (n == 135) {chipPattern = [three,three,one,three];}
if (n == 136) {chipPattern = [zero,one,two,three];}
if (n == 137) {chipPattern = [one,one,two,three];}
...
```

Drawing the geodesics to create the tiles themselves was also done in a rudimentary fashion. I listed all of the necessary coordinates to draw the geodesics required. I did try to use some loops to do this in a much nicer way but could not quite figure out how to get it to correspond with how I had set up the way things are drawn in the earlier work. Thus the a_1 tiles - those adjacent to the centre - for example, are drawn like so:

```
// drawing a1 tiles
if (chipPattern[1]) {
  var a1X =
   [-0.414214,0.414214,0.414214,2.414214,2.414214,-2.414214,
   -2.414214,-0.414214];
  var a1Y = [0.000001,0.000001,0.000001,0.000001,
   0.000001,0.000001,0.000001,0.000001];

  var a1SemiCentreX = [720,720,303.089971,303.089971,
   720,720,1136.910028,1136.910028];
  var a1SemiCentreY =
   [-72.977136,-72.977136,343.933515,343.933515,
   760.843439,760.843439,343.933516,343.933516];
```

```
8   var a1SemiRadius =
     [294.800440,294.800440,294.799818,294.799818,
     294.799925,294.799925,294.799817,294.799817];
9
10  for (var index = 0; index < a1X.length; index += 2) {
11    strokeWeight(2);
12    stroke('black');
13    fill(chipPattern[1]);
14    if (a1X[index] && a1Y[index] && a1X[index + 1] && a1Y[
     index + 1]) {
15       circle(a1SemiCentreX[index], a1SemiCentreY[index], 2 *
     a1SemiRadius[index]);
16    }
17  }
18 }
```

I know this is not the best way and some better coding would make it easier to scale up to tilings that use a lot more chips and go well beyond a distance of three from the centre. However I did not manage to get anything that worked consistently and thought that since what I do have works correctly it will have to do for now.

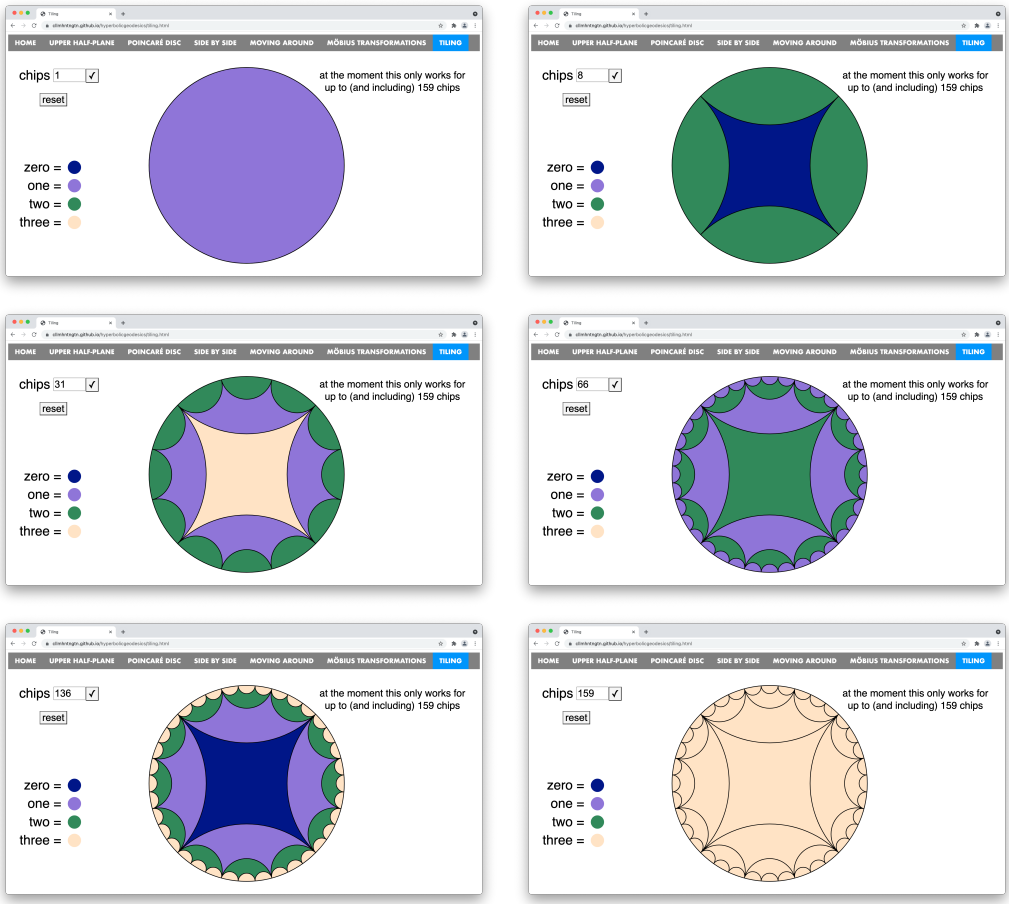


Figure A.7: Examples of coloured tilings of the Poincaré disc when dropping different numbers of chips on the central vertex of the Cayley graph dual to the tiling and letting the model stabilise.